

A Scalable Framework for Enhancing Satellite-Derived Daily Precipitation: Adjusting Values, Aligning Distributions, and Preserving Extremes

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Abstract

Accurate daily precipitation estimates are essential for hydrological modeling, climate risk assessment, and disaster preparedness. Satellite-based precipitation products such as the Integrated Multi-satellitE Retrievals for Global Precipitation Measurement (IMERG) provide global coverage but exhibit systematic biases in daily accumulations, particularly in representing extreme precipitation events. This study presents a hybrid bias correction framework (LSEQM+DL) for daily satellite precipitation that sequentially integrates Linear Scaling (LS) for mean bias removal, Empirical Quantile Mapping (EQM) with Generalized Pareto Distribution (GPD) tail adjustment for distributional alignment, and a Convolutional Neural Network (CNN) refinement that selectively targets extreme precipitation pixels. A station density confidence mask modulates the deep learning influence based on gauge network density, so that the CNN refinement is strongest where the reference dataset (CPC Unified Gauge-Based Analysis of Daily Precipitation, CPC-UNI) is best constrained by observations. Applied to daily IMERG Late Run precipitation over Indonesia (2001-2025) and evaluated against CPC-UNI and 171 independent station observations from Indonesia's Meteorological, Climatological, and Geophysical Agency (BMKG), the framework is assessed through three evaluation pillars aligned with its design objectives: adjusting values, aligning distributions, and preserving extremes. At independent BMKG station locations, the hybrid correction brings the standard deviation ratio from 0.71 (LS) to 1.00 (LSEQM+DL, toward unity), the relative bias from -11.4% to -0.6% , and the 99th-percentile ratio from 0.71 to 1.01, indicating convergence of the corrected distribution toward the observed precipitation statistics. The wet-day frequency ratio improves from 1.21 (LS) to 0.95 (LSEQM+DL), reducing a 21% overestimation of wet-day occurrence to within 5% of observed. These improvements come at a known cost: event detection (POD) decreases from 0.78 to 0.65, reflecting the trade-off inherent to distributional correction methods that adjust marginal statistics without modifying temporal sequencing. Pixel-level temporal accuracy metrics (correlation, root mean square error, Nash-Sutcliffe efficiency) remain largely unchanged, confirming that the framework improves statistical properties rather than day-to-day prediction skill. The World Meteorological Organization (WMO) multi-threshold verification at station locations shows maintained detection skill at rainfall thresholds up to 100 mm/day. Because the approach relies on globally available satellite and gauge-analysis datasets, it is applicable to other tropical and subtropical regions with sparse gauge networks, though recalibration of CNN hyperparameters and GPD thresholds may be required for different climate regimes.

Keywords: Bias Correction, Satellite Precipitation, IMERG, Daily Precipitation, Extreme Events, Deep Learning, Quantile Mapping, Generalized Pareto Distribution, Quality Assessment, Indonesia

1 Introduction

Accurate and timely precipitation estimates are critical for a wide range of scientific and societal applications, including flood forecasting, drought monitoring, hydrological modeling, and climate risk assessments. Satellite-derived precipitation products, such as those from the Global Precipitation Measurement (GPM) mission [1,2], have greatly expanded the global coverage of precipitation observations, especially over regions with sparse ground station networks. However, despite their transformative role, satellite-based estimates often suffer from systematic biases, spatial inconsistencies, and challenges in capturing the full spectrum of precipitation variability [3], particularly for extreme rainfall events. These limitations reduce the reliability of satellite data for operational and scientific applications that depend on an accurate depiction of both mean conditions and high-impact extremes.

A wide range of bias correction techniques has been developed to address these challenges. Traditional statistical methods, such as Linear Scaling (LS), effectively correct mean biases by adjusting the overall magnitude of satellite estimates to align with ground-based observations [4]. Empirical Quantile Mapping (EQM) extends this correction by aligning the entire distribution of satellite precipitation with the reference distribution, adjusting not only the mean but also higher-order statistical moments [5,6]. More recently, methods such as Generalized Pareto Distribution (GPD) fitting have been introduced to specifically address biases in the tails of precipitation distributions, aiming to better preserve extreme events [7]. In parallel, advances in machine learning (ML) and deep learning (DL) have opened new possibilities for bias correction, enabling more flexible and nonlinear mappings between satellite and reference data [8]. Convolutional Neural Networks (CNNs) provide powerful capabilities for modeling complex spatial patterns that traditional statistical methods may overlook [9,10].

Despite these advances, several critical gaps remain. Traditional bias correction techniques, while effective in adjusting mean and moderate rainfall distributions, often underperform in preserving the frequency and magnitude of extreme precipitation events [8,11], precisely the events most relevant for disaster risk management and climate adaptation. Furthermore, many machine learning approaches lack integration with physically interpretable statistical corrections, which can lead to potential issues such as overfitting, instability, or a lack of transparency [12]. There remains a need for a hybrid approach that simultaneously addresses mean biases, distributional alignment, and extreme event preservation in a scalable and reproducible manner.

In this study, we propose a hybrid bias correction framework that sequentially integrates Linear Scaling, Empirical Quantile Mapping with gamma-based distribution fitting and GPD tail adjustment, and a Deep Learning refinement step using Convolutional Neural Networks. The framework is designed to improve daily satellite precipitation estimates by systematically adjusting mean values, aligning distributions, and preserving extremes. Using globally available datasets -- the Integrated Multi-satellitE Retrievals for GPM (IMERG) Late Run (IMERG-L) satellite precipitation [2] and the CPC Global Unified Gauge-Based Analysis of Daily Precipitation (CPC-UNI) gridded station observations [13] -- the methodology is demonstrated over the Indonesian archipelago, selected for its vulnerability to extreme rainfall and complex topographic and climatic conditions. Correction quality is evaluated through 31 verification metrics organized into a composite Continuous Quality Index (CQI), providing a structured assessment across basic statistical, distributional, and temporal dimensions [14,15].

To address the spatial heterogeneity inherent in gauge-based reference datasets, the framework incorporates a station density confidence mask that modulates the influence of the deep learning component based on gauge network density. In regions where the CPC-UNI reference is supported by dense station coverage, the CNN refinement is applied with greater confidence. In station-sparse

regions, where the reference dataset relies on interpolation and may not accurately capture local precipitation characteristics, the framework reverts toward purely statistical corrections. This design provides a systematic approach for managing reference dataset uncertainty in operational bias correction over regions with heterogeneous observational networks.

2 Conceptual Framework and Theoretical Background

Improving satellite-derived precipitation estimates requires addressing biases across multiple statistical dimensions: the mean, the distributional structure, and the representation of extremes. Because these biases are multi-dimensional, this study employs a hybrid framework that sequentially integrates physical-statistical correction methods with machine learning refinement, targeting improvements across the entire precipitation spectrum.

Rationale for a Hybrid Approach

Single-method bias correction strategies often target specific aspects of the precipitation distribution but leave other sources of error unaddressed. Linear Scaling (LS) methods are effective at correcting systematic mean biases but do not modify variance or higher-order moments, leaving distributional mismatches largely unresolved [4,16]. Empirical Quantile Mapping (EQM) extends corrections to the full distribution, aligning the cumulative distribution functions (CDFs) between satellite and reference datasets [5,6]. However, EQM may underestimate or misrepresent the upper extremes, where observational data sparsity and fitting uncertainty are most pronounced [8].

Extreme events, which are often underrepresented in station-based datasets and distorted in satellite retrievals, require specialized treatment. Generalized Pareto Distribution (GPD) fitting provides a statistical mechanism to model precipitation behavior beyond high thresholds, complementing EQM by refining the upper tail of the distribution [7]. Nevertheless, statistical methods alone may struggle to capture complex spatial structures and local-scale anomalies that influence extreme precipitation, particularly over heterogeneous landscapes.

Recent advances in deep learning, particularly the application of Convolutional Neural Networks (CNNs), offer a complementary tool for bias correction. CNNs can learn nonlinear relationships between satellite estimates and reference data, exploiting spatial correlations and subtle patterns not easily captured by traditional techniques [9,17]. Integrating deep learning refinement after statistical corrections allows the framework to adaptively correct localized errors without undermining physical plausibility [10].

Thus, a sequential hybrid approach, starting with mean correction (LS), progressing to distributional correction (EQM with GPD adjustment), and culminating in spatially aware deep learning refinement, is proposed to systematically address all facets of satellite precipitation bias. A critical additional consideration is the spatial heterogeneity of reference dataset quality: the CPC-UNI product, while globally available, relies on interpolation in station-sparse regions and may not accurately represent local extremes [13,18]. The framework addresses this through a station density confidence mask that modulates the deep learning influence based on gauge availability, providing an explicit mechanism for managing training target uncertainty.

The sequential structure of the proposed methodology is illustrated in Figure 1.

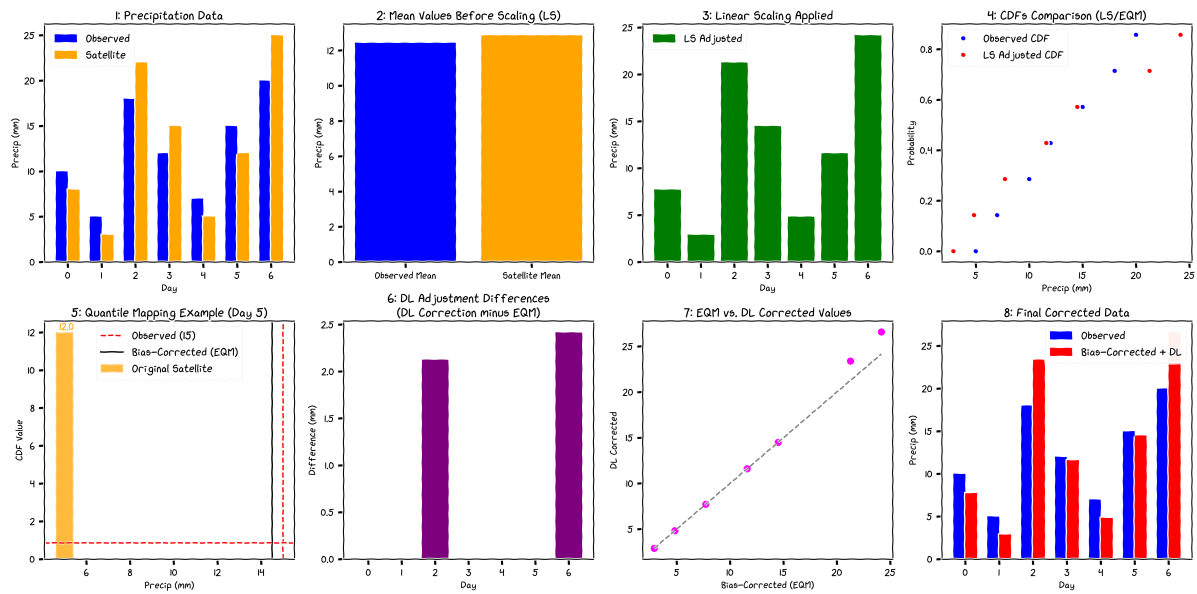


Figure 1. Conceptual workflow of the LSEQM+DL hybrid bias correction framework.

The framework integrates sequential adjustments through LS for mean correction, EQM with GPD tail adjustment for distributional alignment, Deep Learning (DL) refinement targeting extreme precipitation preservation, and station density confidence weighting to manage reference dataset uncertainty.

3 Methods

Bias correction of satellite-derived precipitation products requires not only robust methodological design but also careful data preparation and study site selection. This chapter outlines the geographical focus of the study, the datasets used, and the preprocessing steps taken to ensure consistency and comparability between satellite and reference observations. Each step is tailored to support the development of a globally scalable bias correction framework that enhances precipitation estimates across varying climatic and topographical conditions.

3.1 Study Region

Indonesia, the world's largest archipelagic country, presents a complex and challenging environment for precipitation estimation. Located between the Indian and Pacific Oceans and straddling the equator, Indonesia experiences a tropical climate characterized by high spatial and temporal variability in rainfall [19,20]. Topographical diversity, ranging from lowland plains to mountainous regions exceeding 4,000 meters in elevation, further complicates precipitation patterns. The country is highly vulnerable to extreme rainfall events that frequently trigger flooding, landslides, and other hydrometeorological disasters.

This study adopts Indonesia as the area of interest (AOI) due to its critical need for improved precipitation monitoring and its representative challenges for satellite-based hydrometeorological applications. While the Indonesia's Meteorological, Climatological, and Geophysical Agency (BMKG) station data used for validation are specific to Indonesia, the primary datasets utilized for bias correction, IMERG and CPC-UNI, are globally available. Moreover, the hybrid bias correction methodology developed here is designed to be scalable and replicable across other regions worldwide, supporting broader applications in climate research, disaster risk reduction, and hydrological modeling beyond Indonesia.

3.2 Data and Preprocessing

Developing a reliable bias correction framework demands the careful selection of datasets that represent both satellite-based estimates and ground-based observations [22]. This section describes the data sources utilized and outlines the preprocessing steps undertaken to harmonize their spatial and temporal characteristics.

3.2.1 IMERG Satellite Precipitation Data

The Integrated Multi-satellitE Retrievals for GPM (IMERG) Late Run (IMERG-L) product is selected as the satellite-derived precipitation dataset for this study [2,21]. IMERG-L offers an optimal balance between latency and data quality, providing near-real-time precipitation estimates with a delay of approximately 14 hours. Unlike the IMERG Final Run, which incorporates station data into its post-processing, the Late Run product relies purely on satellite observations, making it ideal for evaluating bias correction methods without the influence of gauge data assimilation. IMERG-L data are available at a high spatial resolution of 0.1 degrees (~10 km) and are directly accessed from the official NASA Goddard Earth Sciences Data and Information Services Center (GES DISC).

3.2.2 IMERG Final Run for Complementary Evaluation

In addition to the IMERG-L data used for bias correction, this study also utilizes the IMERG Final Run (IMERG-F) product as a complementary evaluation dataset. IMERG-F incorporates gauge-based observations during its post-processing, aiming to further improve precipitation accuracy relative to purely satellite-derived estimates. Although it benefits from station assimilation, the IMERG-F is not used in the bias correction procedure itself to maintain independence between correction inputs and validation references. Instead, the IMERG-F is employed for comparative analysis, providing an additional benchmark to assess the effectiveness of the bias correction applied to the IMERG-L data. IMERG-F data are also available at a 0.1-degree spatial resolution and were obtained directly from the NASA GES DISC.

3.2.3 CPC-UNI Gridded Observational Data

The Climate Prediction Center Unified Gauge-Based Analysis of Global Daily Precipitation (CPC-UNI) dataset is employed as the reference observation for bias correction [13,18]. CPC-UNI offers the highest-resolution global daily gridded precipitation product derived from station measurements, available at a 0.5-degree spatial resolution (~50 km). Its broad spatial coverage and gauge-based foundation make it suitable for large-scale satellite precipitation correction efforts. However, the gridded values in CPC-UNI are derived through interpolation, and the density of contributing stations varies geographically. In regions with sparse station networks, including parts of Indonesia, interpolated values may fail to accurately capture local extremes, particularly in complex terrain. This limitation highlights the need to integrate CPC-UNI with satellite-derived data and apply additional distributional and extreme-event corrections. The CPC-UNI data used in this study were obtained directly from the NOAA Physical Sciences Laboratory.

3.2.4 Independent Validation Data

Independent validation is conducted using ground-based observations from Indonesia's Meteorological, Climatological, and Geophysical Agency (BMKG). BMKG operates an extensive network of weather stations providing daily precipitation measurements across the Indonesian archipelago. These independent station data are not used in the bias correction process itself but serve to evaluate the performance of the corrected IMERG precipitation estimates. Validation focuses on key criteria, including the correct detection of rainfall events, the ability to represent observed extremes, and the assessment of spatial consistency between corrected satellite estimates

and ground truth. Independent validation data provide an external assessment of correction quality that is not influenced by the CPC-UNI training reference, particularly for extreme rainfall conditions relevant to disaster risk management. BMKG daily precipitation records were obtained directly from the agency's online data repository.

3.2.5 Preprocessing and Harmonization

To ensure comparability between the satellite-derived and ground-based datasets, several preprocessing steps were undertaken.

- **Spatial harmonization.** The CPC-UNI dataset is available at 0.5° spatial resolution, five times coarser than the 0.1° IMERG grid. To enable pixel-level correction, CPC-UNI is regridded to the IMERG grid using nearest-neighbor interpolation, which preserves the discrete gauge-analyzed values without artificial smoothing [13]. This regridded product is used for daily time-series pairing during the correction loop.
To preserve the statistical independence of the gauge-based reference during distribution fitting, the framework retains the CPC-UNI data at its native 0.5° resolution alongside the regridded product. Following the Bias Correction Spatial Disaggregation (BCSD) principle [23], distribution parameters and climatological means are estimated at the native 0.5° CPC-UNI resolution, then bilinearly interpolated to the 0.1° IMERG grid before quantile mapping is applied. This produces spatially smooth correction fields while preserving the statistical integrity of the gauge-based reference. The approach is described in detail in Section 3.3.
- **Temporal alignment.** Daily time steps were synchronized across all datasets, and a strictly monotonic, duplicate-free time axis was enforced. Where duplicate time stamps existed due to data processing artifacts, only the first occurrence was retained.
- **Land-sea masking.** A binary land-sea mask derived from official land boundary datasets was applied to both IMERG and CPC-UNI at data loading, setting all ocean grid cells to missing values. This prevents ocean zeros from contaminating statistical calculations (e.g., mean bias, percentile estimation) that rely on the assumption of land-based precipitation sampling.
- **Wet-day threshold.** A wet-day threshold of 1.0 mm/day is applied throughout the framework for categorical metric computation and precipitation day classification, following World Meteorological Organization (WMO) recommendations for operational precipitation monitoring in tropical regions where trace precipitation is frequent [24].
- **Latitude orientation.** The CPC-UNI dataset uses descending latitude ordering. To ensure consistent array operations, latitude coordinates are checked and reindexed to ascending order where necessary prior to any spatial alignment or computation.

These preprocessing steps produce spatially and temporally consistent datasets, enabling robust application of bias correction techniques and minimizing errors due to resolution or coverage discrepancies.

3.3 Hybrid Bias Correction Methodology

Building upon the conceptual framework introduced in Chapter 2, the operational implementation follows a sequential four-stage structure. Each processing stage targets a specific source of bias in satellite-derived daily precipitation estimates. All bias correction steps are performed separately for each dekadal period (three subdivisions per month: days 1--10, 11--20, and 21 to end of month). Within each dekad, daily precipitation values are pooled across all available years (2001--2025), yielding approximately 250 samples per dekad. This multi-year pooling strategy enhances the statistical robustness of distribution fitting and extreme value estimation, while preserving the seasonal characteristics of each dekadal window.

The overall hybrid bias correction workflow is illustrated in Figure 2.

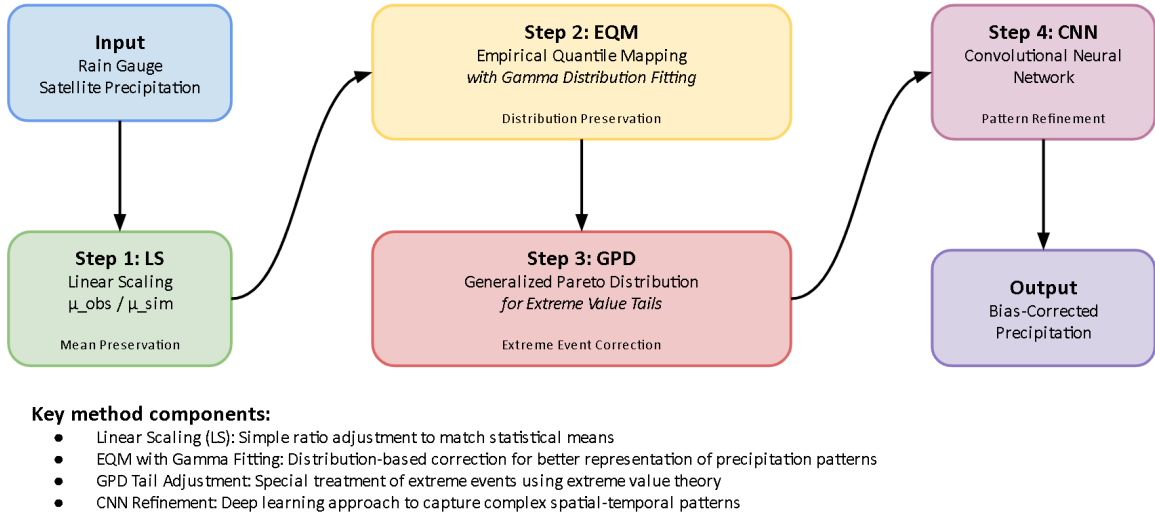


Figure 2. Overview of the hybrid bias correction workflow integrating Linear Scaling (LS), Empirical Quantile Mapping (EQM) with Gamma fitting, Generalized Pareto Distribution (GPD) tail adjustment, and Convolutional Neural Network (CNN) refinement.

Stage 1: Linear Scaling

Linear Scaling corrects the systematic mean bias by applying a multiplicative factor derived from the ratio of reference to satellite mean precipitation. For each dekad, the scaling factor C_{LS} is computed as:

$$C_{LS} = \frac{\bar{P}_{ref}}{\bar{P}_{sat}}$$

where $\bar{P}_{ref}$ and $\bar{P}_{sat}$ are the mean daily precipitation of the CPC-UNI reference and IMERG-L satellite datasets, respectively, pooled over the same dekadal period across all years. The corrected precipitation is:

$$P_{LS} = P_{sat} \cdot C_{LS}$$

Grid cells where the satellite mean is zero are left unchanged (scaling factor set to unity). Following the dual-resolution strategy described in Section 3.2, the CPC-UNI dekadal mean $\bar{P}_{ref}$ is computed at the native 0.5° grid and bilinearly interpolated to 0.1° before computing C_{LS} .

This produces a spatially smooth scaling field that transitions gradually across CPC-UNI cell boundaries, rather than exhibiting the stepwise jumps that would result from computing the mean on the regridded product. This step aligns the first moment of the precipitation distribution prior to distributional corrections.

Stage 2: Empirical Quantile Mapping with Gamma Distribution Fitting (EQM)

Empirical Quantile Mapping aligns the full cumulative distribution function (CDF) of the LS-corrected satellite precipitation to that of the reference observations. To improve the stability and smoothness of the CDFs, particularly in regions with limited sample sizes, gamma distributions are fitted to the precipitation data prior to mapping. Gamma parameters (shape k and scale θ) are estimated using maximum likelihood estimation (MLE) with the location parameter fixed at zero to ensure physical validity of non-negative precipitation values [15]. The quantile correction for each satellite precipitation value P_{LS} is performed by:

$$P_{EQM} = F_{ref}^{-1}(F_{sat}(P_{LS}))$$

where F_{sat} and F_{ref} denote the fitted gamma CDFs of the satellite and reference datasets, respectively, and F_{ref}^{-1} is the inverse CDF (quantile function) of the reference distribution.

The IMERG-side gamma CDF (F_{sat}) is fitted independently at each 0.1° pixel, preserving the full spatial detail of the satellite product. The CPC-UNI-side gamma CDF (F_{ref}) is fitted at the native 0.5° resolution, where each cell contains a statistically independent time series of gauge-analyzed values. The resulting gamma parameters (shape k_{ref} and scale θ_{ref}) are then bilinearly interpolated to the 0.1° IMERG grid.

This dual-resolution fitting strategy follows the BCSD principle [23]: by fitting at the resolution where independent information exists and spatially disaggregating the fitted parameters, the quantile transfer function varies smoothly across the domain while each parameter estimate draws on a genuinely distinct observational sample.

Dry-day handling. Because daily tropical precipitation is a mixed discrete-continuous variable with a large atom at zero, the gamma distribution is fitted only on wet days ($P \geq 1$ mm/day) rather than on the full sample. Fitting a gamma to the mixed sample would bias the shape parameter toward zero and distort the mapped quantiles. The dry-day fraction is handled explicitly at the mapping stage following the unconditional-CDF construction of Cannon et al. [11]. Let p_{dry}^{sat} and p_{dry}^{ref} denote the satellite and reference dry-day probabilities estimated at each pixel from the empirical sample. For a satellite wet-day value $P_{LS} \geq 1$ mm/day, the unconditional quantile position is $y = p_{dry}^{sat} + (1 - p_{dry}^{sat}) \cdot F_{sat}^{wet}(P_{LS})$, where F_{sat}^{wet} is the gamma CDF fitted on wet values only. The reference quantile is then obtained from the analogous reference unconditional CDF, with values falling below p_{dry}^{ref} mapped to zero. This construction preserves the reference dry-day frequency by design and prevents satellite drizzle from being mapped to spurious low-intensity wet-day values.

Stage 3: Generalized Pareto Distribution (GPD) Tail Adjustment

Extreme precipitation values are modeled separately to preserve the frequency and magnitude of high-impact events that gamma-based EQM may underrepresent. Precipitation exceedances above the 80th percentile of the wet-day sample are fitted using the Generalized Pareto Distribution, with separate thresholds u_{sat} and u_{ref} estimated from the satellite and reference wet-value distributions, respectively. The 80th percentile threshold was selected to balance two competing requirements: a sufficiently large number of exceedances for stable GPD parameter estimation, and a threshold high enough to isolate genuinely extreme behavior from the bulk distribution [7]. This threshold is consistent with recommendations for tropical regions with frequent heavy rainfall, where lower thresholds risk contaminating GPD fits with moderate rainfall, while higher thresholds yield too few exceedances for reliable estimation [25].

For precipitation values exceeding threshold u , the exceedances $y = P - u$ are modeled using the GPD probability density function:

$$f(y) = \frac{1}{\sigma} \left(1 + \xi \frac{y}{\sigma}\right)^{-(1+1/\xi)}$$

where ξ is the shape parameter and σ is the scale parameter. The location parameter is fixed at zero because exceedances are non-negative by construction; leaving the location free would bias the fitted tail [7]. To improve robustness against data sparsity, GPD parameters are estimated using K -fold cross-validation ($K = 5$) with shuffled folds, and the final parameters are averaged across all folds to reduce sensitivity to individual outliers [25]. Conditional probability mapping is then applied to correct extreme quantiles. For a satellite value $P_{sat} > u_{sat}$, the conditional exceedance probability is computed on the satellite side using its own threshold,

$$p_c = \frac{F_{sat}(P_{sat}) - F_{sat}(u_{sat})}{1 - F_{sat}(u_{sat})},$$

and the GPD-corrected value is obtained from the reference GPD as

$$P_{LSEQM} = u_{ref} + G_{ref}^{-1}(p_c),$$

where G_{ref}^{-1} is the inverse GPD fitted to the reference exceedances and u_{ref} is the reference wet-value 80th percentile. Computing p_c on the satellite side ensures that the tail-substitution probability chain is internally consistent with the unconditional gamma CDF used in the body of the distribution, and maintains continuity at the gamma-GPD boundary.

As with the gamma parameters, the GPD threshold, shape, and scale parameters for the CPC-UNI reference are fitted at the native 0.5° resolution and bilinearly interpolated to the 0.1° grid. The 99.9th percentile cap is similarly computed at 0.5° and interpolated, maintaining spatial consistency across all correction stages.

After GPD adjustment, corrected values are capped at the interpolated 99.9th percentile of the reference distribution to prevent physically unrealistic extremes arising from GPD extrapolation.

Stage 4: Deep Learning Refinement (CNN)

Although the preceding statistical corrections substantially reduce biases in mean, distribution, and extremes, residual errors in extreme precipitation pixels often persist due to complex local topographic and atmospheric processes. To address these remaining biases, a Convolutional Neural Network (CNN) is employed to refine the corrected precipitation fields, with its corrections selectively applied to extreme pixels through alpha-blending.

The CNN is trained separately for each dekadal period, using the LSEQM-corrected fields as input and CPC-UNI observations as the training target. This two-stage design means the CNN receives input from the same statistical domain at both training and inference time, avoiding the domain shift problem that arises when training directly on raw satellite data [9].

Each daily precipitation field is independently normalized to $[0, 1]$ by dividing by its maximum value, allowing the CNN to focus on spatial pattern learning rather than absolute intensity differences:

$$\hat{P}_i = \frac{P_i}{\max(P_i)}$$

The CNN architecture consists of:

- Two convolutional layers (3×3 filters) with 32 and 64 filters, respectively, using ReLU activation and same-padding;
- Max pooling (2×2) and dropout regularization after each convolutional layer (rates of 0.2 and 0.3);
- A flattening layer followed by a dense layer (128 units, ReLU activation) with dropout (rate 0.4);
- A linear output layer reconstructing the corrected precipitation field.

The model is trained by minimizing the mean squared error (MSE) loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left(\hat{P}_{DL,i} - \hat{P}_{ref,i} \right)^2$$

using the Adam optimizer [26], with an 80/20 training-validation split. Early stopping with a patience of 5 epochs prevents overfitting, and a batch size of 64 is used for gradient updates.

The schematic of the CNN refinement architecture, including the alpha-blending integration, is provided in Figure 3.

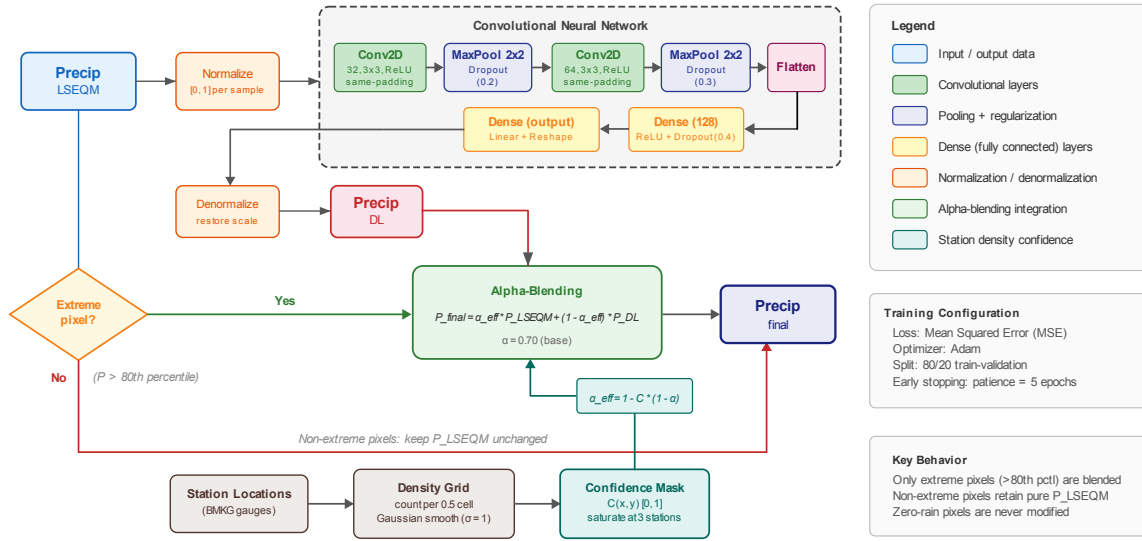


Figure 3. CNN refinement pipeline: LSEQM-corrected daily fields are normalized, passed through two convolutional blocks and a Flatten-Dense bottleneck, and reconstructed as a refined field. The refined field is then combined with the LSEQM field through the station-density-modulated alpha-blending step applied only to pixels exceeding the 80th percentile.

Alpha-Blending Integration

The CNN predictions are not applied uniformly across all grid cells. Instead, an alpha-blending strategy selectively integrates the DL refinement with the statistical corrections. For each grid cell and time step, the final corrected precipitation is:

$$P_{final} = \alpha_{eff} \cdot P_{LSEQM} + (1 - \alpha_{eff}) \cdot P_{DL}$$

where α_{eff} is the effective blending weight. This blending is applied only to pixels identified as extreme, defined as those where P_{LSEQM} exceeds the 80th percentile of the pixel's dekadal distribution.

Non-extreme pixels retain their LSEQM values without modification, and pixels with zero precipitation are preserved as zero. The base blending weight $\alpha = 0.70$ assigns 70% weight to the physical-statistical correction and 30% to the DL refinement.

Station Density Confidence Weighting

The reliability of the CPC-UNI reference varies spatially depending on the density of contributing gauge stations. To account for this heterogeneity, a station density confidence mask modulates the blending parameter. Gauge station counts per CPC-UNI native grid cell (0.5°) are computed from the BMKG station network, Gaussian-smoothed ($\sigma = 1.0$ grid cells) to prevent sharp boundaries, then normalized to $[0, 1]$ by dividing by a saturation count of 3 stations per cell. The confidence mask $C(x, y)$ modulates the blending parameter as:

$$\alpha_{eff}(x, y) = 1.0 - C(x, y) \cdot (1.0 - \alpha)$$

This formulation provides a smooth spatial transition: station-dense areas receive the full DL contribution ($\alpha_{eff} = 0.70$), while station-sparse areas revert toward pure LSEQM ($\alpha_{eff} \rightarrow 1.0$). This approach reduces the risk of the DL model introducing artifacts in regions where the training target itself is uncertain [27].

3.4 Model Evaluation and Quality Assessment

Evaluating bias correction of daily precipitation requires a multi-metric framework that captures statistical accuracy, distributional consistency, temporal pattern preservation, and the representation of extreme events. Daily precipitation exhibits high temporal variability, frequent dry periods, and intermittent extreme events, making it more difficult to evaluate than monthly-aggregated precipitation. A single performance metric cannot capture correction quality across all relevant dimensions --- event timing, intensity distribution, dry spell behavior, and extremes --- and this study therefore employs a structured suite of statistical, categorical, temporal, and distributional metrics combined with integrated quality indices, consistent with recommended practice for satellite precipitation verification [14,15].

3.4.1 Nine-Combination Evaluation Design

The evaluation follows a systematic 3 x 3 factorial design (Table 1). Three reference datasets are used: (1) CPC-UNI, the gauge-based product used as the training reference for bias correction; (2) IMERG-L, the uncorrected satellite product; and (3) IMERG-F, the gauge-adjusted IMERG Final Run product, which serves as an independent benchmark. Each reference is compared against three correction stages: LS (mean correction only), LSEQM (mean + distributional correction), and LSEQM+DL (full hybrid correction). This yields nine reference-test combinations, enabling three complementary perspectives:

- **CPC-UNI vs. corrected products:** Measures absolute correction quality against the training reference, assessing how well each stage reproduces the gauge-based observations.
- **IMERG-L vs. corrected products:** Quantifies improvement relative to the uncorrected satellite baseline, isolating the added value of bias correction.
- **IMERG-F vs. corrected products:** Evaluates performance against an independent, gauge-adjusted satellite product that was not used in the correction procedure, providing a less circular validation.

Table 1. Evaluation matrix: nine reference-test combinations.

	LS	LSEQM	LSEQM+DL
CPC-UNI	CPC-UNI vs LS	CPC-UNI vs LSEQM	CPC-UNI vs LSEQM+DL
IMERG-L	IMERG-L vs LS	IMERG-L vs LSEQM	IMERG-L vs LSEQM+DL
IMERG-F	IMERG-F vs LS	IMERG-F vs LSEQM	IMERG-F vs LSEQM+DL

3.4.2 Evaluation Metrics

The performance of each bias correction stage is evaluated using 31 verification metrics organized into four categories: continuous statistical measures, categorical event detection, temporal pattern fidelity, and distributional consistency. All metrics are computed at the pixel level for each dekadal period, enabling spatially explicit assessment of correction quality.

A wet-day threshold of 1.0 mm/day is applied for all categorical metrics, consistent with recommendations for tropical regions where trace precipitation is frequent [24].

Table 2 summarizes the complete metric suite. Formulas for the principal metrics are provided below; standard definitions for the remaining metrics follow established references [14,15].

Table 2. Summary of verification metrics

Category	Metric	Abbreviation	Unit	Perfect Score
Continuous	Relative Bias	RB	ratio	0
	Pearson Correlation	CORR	--	1
	Root Mean Square Error	RMSE	mm/day	0
	Mean Absolute Error	MAE	mm/day	0
	Nash-Sutcliffe Efficiency	NSE	--	1
	Standard Deviation (ref, test)	STDEV	mm/day	equal
	Standard Deviation Ratio	SDR	ratio	1
	KS Test Statistic / <i>p</i> -value	KS	--	0 / 1
Categorical	Probability of Detection	POD	--	1
	False Alarm Ratio	FAR	--	0
	Critical Success Index	CSI	--	1
	Equitable Threat Score	ETS	--	1
Temporal	Frequency of Precipitation Days	FPD	%	equal
	Mean Wet-Day Precipitation	MDWP	mm/day	equal
	Maximum Dry Spell Length	DSL	days	equal
Distributional	Percentiles (Q_{25} -- Q_{99})	Q_{xx}	mm/day	equal

Continuous metrics.

The Relative Bias measures the proportional difference between test and reference totals:

$$RB = \frac{\sum_{i=1}^n P_{test,i} - \sum_{i=1}^n P_{ref,i}}{\sum_{i=1}^n P_{ref,i}}$$

The Pearson Correlation Coefficient quantifies the linear relationship:

$$r = \frac{\sum_{i=1}^n (P_{ref,i} - \bar{P}_{ref})(P_{test,i} - \bar{P}_{test})}{\sqrt{\sum_{i=1}^n (P_{ref,i} - \bar{P}_{ref})^2 \cdot \sum_{i=1}^n (P_{test,i} - \bar{P}_{test})^2}}$$

The Root Mean Square Error and Mean Absolute Error quantify error magnitude:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_{test,i} - P_{ref,i})^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |P_{test,i} - P_{ref,i}|$$

The Nash-Sutcliffe Efficiency provides a normalized measure of prediction skill relative to the reference mean:

$$NSE = 1 - \frac{\sum_{i=1}^n (P_{ref,i} - P_{test,i})^2}{\sum_{i=1}^n (P_{ref,i} - \bar{P}_{ref})^2}$$

NSE values range from $-\infty$ to 1, where values above 0.5 indicate satisfactory performance and values below zero indicate that the reference mean is a better predictor than the corrected product [28,29].

The Standard Deviation is computed independently for both reference and test datasets using the sample estimator (with Bessel's correction, $ddof = 1$):

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (P_i - \bar{P})^2}$$

The Standard Deviation Ratio (SDR) quantifies the relative variability of the corrected product compared to the reference:

$$\text{SDR} = \frac{\sigma_{test}}{\sigma_{ref}}$$

SDR values greater than 1 indicate over-dispersion (excessive variability), while values less than 1 indicate under-dispersion. A perfect correction preserves the reference variability ($\text{SDR} = 1$).

Categorical metrics.

Event detection is assessed using the probability of detection, false alarm ratio, and critical success index, defined in terms of contingency table entries (hits H , misses M , and false alarms F):

$$\text{POD} = \frac{H}{H+M}, \quad \text{FAR} = \frac{F}{H+F}, \quad \text{CSI} = \frac{H}{H+M+F}$$

The Equitable Threat Score (ETS) adjusts CSI by removing hits attributable to random chance:

$$\text{ETS} = \frac{H - H_{random}}{H+M+F-H_{random}}, \quad H_{random} = \frac{(H+M)(H+F)}{N}$$

where N is the total number of observations. $\text{ETS} = 0$ indicates no skill beyond random forecasts; $\text{ETS} = 1$ indicates perfect detection [15].

Temporal metrics.

Three metrics characterize the temporal precipitation patterns. The Frequency of Precipitation Days (FPD) measures the percentage of days exceeding the wet-day threshold, computed separately for reference and test datasets:

$$\text{FPD} = \frac{N_{wet}}{N_{valid}} \times 100$$

where N_{wet} is the number of days with precipitation ≥ 1.0 mm and N_{valid} is the total number of valid (non-missing) days. Comparing FPD between reference and test reveals whether the correction alters the wet-day frequency.

The Mean Wet-Day Precipitation (MDWP) captures the average intensity on days that exceed the wet-day threshold:

$$\text{MDWP} = \frac{1}{N_{wet}} \sum_{i=1}^{N_{wet}} P_i \quad \text{for } P_i \geq 1.0 \text{ mm/day}$$

The Maximum Dry Spell Length (DSL) is defined as the longest consecutive run of days below the wet-day threshold within the evaluation period. It is computed as:

$$\text{DSL} = \max_j \{L_j\}$$

where L_j is the length of the j -th consecutive dry spell. This metric evaluates whether the bias correction preserves realistic dry spell persistence, which is relevant for drought monitoring and agricultural applications.

Distributional metrics.

Six percentiles ($Q_{25}, Q_{50}, Q_{75}, Q_{90}, Q_{95}, Q_{99}$) are computed for both reference and test datasets to evaluate distributional agreement across the full precipitation spectrum, with particular emphasis on the upper tail (Q_{90} and above) relevant to extreme event assessment. The two-sample Kolmogorov-Smirnov test evaluates whether the corrected and reference samples originate from the same underlying distribution:

$$D_{KS} = \sup_x |F_{test}(x) - F_{ref}(x)|$$

Metrics are computed in two modes: (1) a timeseries mode producing one metric grid per year, enabling temporal trend analysis of correction quality; and (2) a single-dekad mode pooling all years into one aggregated assessment.

3.4.3 Composite Quality Assessment Framework

While individual metrics provide insight into specific aspects of correction performance, a composite framework is needed to evaluate the overall quality of bias-corrected products across multiple dimensions simultaneously. The Continuous Quality Index (CQI) aggregates multiple metrics into three principal dimensions: basic statistical quality, distributional quality, and temporal pattern quality.

Basic Statistical Quality Score (BSQS)

The basic statistical quality score combines three continuous metrics:

$$BSQS = 0.30 \cdot S_{RB} + 0.30 \cdot S_{RMSE} + 0.40 \cdot S_{NSE}$$

where $S_{RB} = 1 - \min(|RB|, 1)$ penalizes both over- and under-estimation, $S_{RMSE} = \exp(-RMSE/5)$ applies exponential decay normalization, and $S_{NSE} = \max(\min(NSE, 1), 0)$ clips NSE to $[0, 1]$. The Pearson Correlation Coefficient is not included as a separate component because it is implicitly captured within the NSE formulation; including both would introduce double-counting of linear association [30]. NSE receives a higher weight (0.40) as it simultaneously accounts for bias, variance, and correlation.

Distribution Quality Score (DQS)

The distribution quality score evaluates the preservation of precipitation intensity distributions across both moderate and extreme regimes:

$$DQS = 0.45 \cdot S_{extreme} + 0.25 \cdot S_{general} + 0.15 \cdot S_{var} + 0.15 \cdot S_{KS}$$

where $S_{extreme}$ is the weighted score across the 90th, 95th, and 99th percentile differences (sub-weights 0.30, 0.30, and 0.40), $S_{general}$ captures the 25th, 50th, and 75th percentile differences (sub-weights 0.30, 0.40, 0.30), and S_{var} measures the relative error in the standard deviation ratio. The term S_{KS} is the Kolmogorov-Smirnov test p -value, providing an overall measure of distributional similarity. The weighting scheme assigns the largest share (0.45) to extreme percentile matching, reflecting the emphasis on preserving high-impact events.

Each percentile score is computed as:

$$S_{Q_p} = 1 - \min\left(\frac{|Q_{p,test} - Q_{p,ref}|}{Q_{p,ref} + 0.1}, 1\right)$$

and the variability score as:

$$S_{var} = 1 - \min(|1 - SDR|, 1)$$

Temporal Quality Score (TQS)

The temporal quality score assesses the corrected dataset's ability to reproduce realistic temporal precipitation patterns:

$$TQS = 0.40 \cdot S_{CSI} + 0.30 \cdot S_{event} + 0.30 \cdot S_{spell}$$

where S_{CSI} is the Critical Success Index, providing a balanced joint measure of detection and false alarm performance; S_{event} combines Probability of Detection and False Alarm Ratio:

$$S_{event} = 0.6 \cdot POD + 0.4 \cdot (1 - FAR)$$

and S_{spell} evaluates the normalized error in maximum dry spell length:

$$S_{spell} = 1 - \min \left(\frac{|DSL_{test} - DSL_{ref}|}{DSL_{ref} + \epsilon}, 1 \right)$$

where $\epsilon = 10^{-6}$ prevents division by zero.

Overall Continuous Quality Index (CQI)

The CQI synthesizes the three component scores into a single normalized indicator:

$$CQI = 0.35 \cdot BSQS + 0.35 \cdot DQS + 0.30 \cdot TQS$$

Equal emphasis is placed on basic statistical and distributional quality (0.35 each), with slightly lower weight for temporal quality (0.30), reflecting the particular importance of intensity accuracy for climate risk applications. The CQI is classified into four quality categories (Table 3).

Table 3. CQI categorical classification

Class	CQI Range	Interpretation
Excellent	≥ 0.8	Correction meets high-performance standards across all metrics
Good	$[0.6, 0.8)$	Well-performing with minor deficiencies in some areas
Fair	$[0.4, 0.6)$	Basic improvements achieved but moderate performance gaps remain
Poor	< 0.4	Substantial deficiencies and limited reliability

Note: Because the majority of pixels in this study fall within the Fair range, Figure 6b employs a finer six-tier subdivision (Poor, Marginal, Fair-Low, Fair-High, Good, Excellent) to reveal spatial structure within the dominant quality class.

3.4.4 Confidence Assessment

The confidence score provides an additional layer of information indicating the reliability of the computed quality metrics. Because the statistical stability of pixel-level metrics depends on sample size, metric consistency, and distributional agreement, the confidence computation is adapted to the evaluation mode.

For the aggregated (single-dekad) evaluation, where sample sizes are large by design (~ 230 daily values pooled across all years):

$$C = 0.60 \cdot S_{consistency} + 0.40 \cdot S_{dist}$$

For the timeseries evaluation, where sample sizes vary per year (~ 10 daily values per dekad):

$$C = 0.40 \cdot S_{sample} + 0.30 \cdot S_{dist} + 0.30 \cdot S_{consistency}$$

where S_{sample} is the fraction of valid (non-missing) time steps, $S_{consistency}$ measures agreement among normalized basic metrics computed as:

$$S_{consistency} = 1 - \text{std} \left(\tilde{S}_{RB}, \tilde{S}_{NSE}, \text{POD}, 1 - \text{FAR} \right)$$

where $\tilde{S}$ denotes the normalized score (mapped to $[0, 1]$ where 1 is optimal), and S_{dist} is the Kolmogorov-Smirnov test p -value (higher p -values indicate greater distributional similarity and thus higher confidence).

Grid cells or time periods with low confidence scores are flagged, highlighting regions where the evaluation may be less reliable due to data limitations or inconsistent correction behavior.

4 Results and Discussion

Before examining correction performance, it is important to characterize the starting conditions. The uncorrected IMERG Late Run over Indonesia exhibits a systematic wet bias against the CPC-UNI reference, with domain-median relative bias of approximately +15% during the wet season (October-March). This bias has a conditional structure: IMERG overestimates precipitation on days that CPC-UNI classifies as light rain while simultaneously underestimating moderate-to-heavy events (Section 4.1.1), a pattern consistent with published IMERG-L evaluations over the Maritime Continent [31,32,33]. The CPC-UNI reference itself carries important limitations: at 0.5° resolution with 180 contributing stations across Indonesia, gauge density is heavily concentrated in Java and southern Sumatra (Section 4.6, Figure 10a), leaving eastern Indonesia (Papua, Maluku, Nusa Tenggara) dependent on interpolation that may not capture local extremes. The independent BMKG station network used for validation (171 stations with ≥ 30 valid days) shares a similar spatial distribution, with station density highest in Java. This geographic concentration means that domain-median validation statistics are more representative of conditions in station-dense western Indonesia than in the data-sparse east --- a caveat that should be borne in mind when interpreting the results below.

4.1 Progressive Improvement Across Correction Stages

The hybrid framework targets three complementary aspects of satellite precipitation bias: adjusting values (mean and magnitude), aligning distributions (statistical shape), and preserving extremes (tail accuracy). To evaluate whether each correction stage delivers on its intended objective, domain-wide verification metrics were computed for the uncorrected IMERG Late Run and the three correction products (LS, LSEQM, and LSEQM+DL) against the CPC-UNI gauge-based reference. Table 4 organizes the evaluation around these three design objectives, supplemented by categorical event detection metrics. All values represent the spatial median across Indonesian land pixels, averaged over 36 dekadal periods (12 months $\times$ 3 dekads per month). Ratio metrics express the corrected product relative to the reference (target = 1.0).

Table 4. Domain-wide evaluation by correction stage against CPC-UNI, organized by the three design objectives of the hybrid framework. Values represent the spatial median across all land pixels, averaged over all 36 dekadal periods. Ratio metrics express test/reference. Bold indicates the value closest to the target.

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Value Adjustment	Relative Bias	0.0000	0.0697	0.0652	0
	Std. Dev. Ratio	0.9707	1.1587	1.1475	1.0

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Distribution Alignment	KS p -value (%)	6.56	0.00	0.00	100
	Wet-Day Freq. Ratio	1.0397	0.9466	0.9462	1.0
	Mean Wet-Day Precip Ratio	0.9611	1.1569	1.1519	1.0
Extreme Preservation	Q_{95} Ratio	0.9459	1.1711	1.1626	1.0
	Q_{99} Ratio	0.9669	1.2117	1.1964	1.0
Event Detection	POD	0.7669	0.7069	0.7069	1.0
	CSI	0.5890	0.5723	0.5724	1.0
	FAR	0.2705	0.2476	0.2475	0
Temporal Skill (acknowledged limitation)	Pearson Corr	0.3430	0.3451	0.3476	1.0
	RMSE (mm/day)	13.1051	14.1797	14.0717	0
	NSE	— 0.2728	— 0.5479	—0.5241	1.0

The results reveal a pattern consistent with the design intent of each correction stage. Linear Scaling drives the domain-wide relative bias against CPC-UNI to zero by construction, since C_{LS} is defined exactly as the ratio of reference to satellite dekadal means; the standard deviation ratio of ≈ 0.97 indicates that a uniform multiplicative rescaling leaves the distributional shape essentially unchanged. The addition of EQM with GPD tail adjustment (LSEQM) then modifies the shape of the distribution: the wet-day-mean ratio moves from 0.96 (LS) to 1.16 and the Q_{95}/Q_{99} ratios move from below unity to above unity (Table 4). These shifts are intentional rather than deficient: LS by construction under-represents wet-day intensity because it also rescales drizzle days upward, and EQM with a wet-day-only gamma fit plus explicit dry-day handling (Section 3.3) lifts the conditional wet-day distribution back toward the reference. The apparent "overshoot" at the upper percentiles against CPC-UNI is analyzed further in Sections 4.5 and 4.8.5, where independent BMKG station validation shows that the LSEQM+DL Q_{99} ratio lands at 1.01 against gauge observations, compared with 0.71 for LS --- the opposite direction from the CPC-UNI-based assessment.

The CNN refinement (LSEQM+DL) provides targeted adjustment at the extreme end: the standard deviation ratio moves slightly closer to unity and Q_{95}/Q_{99} move modestly toward the reference. The CNN effect is intentionally small, reflecting the conservative alpha-blending design ($\alpha = 0.70$ applied only to extreme pixels), which limits the DL contribution to at most 30% of the correction signal and only where the station density confidence mask permits.

Event detection shows a trade-off inherent to distributional correction: POD and CSI shift marginally across correction stages (Table 4), reflecting the fact that quantile mapping adjusts wet-day precipitation magnitudes and can move marginal events above or below the 1 mm/day detection threshold. FAR remains essentially unchanged, indicating that the redistribution of values does not systematically inflate false alarms.

An important characteristic of distributional bias correction methods is that they adjust marginal distributions without modifying the temporal sequencing of satellite estimates [34,11]. Accordingly, pixel-level temporal accuracy metrics show limited change across correction stages: Pearson correlation remains at approximately 0.35, RMSE at approximately 13 mm/day, and NSE remains negative (approximately -0.3) for all methods. These values reflect the inherent difficulty of reproducing gauge-interpolated daily precipitation fields at 0.1° resolution rather than a deficiency of the correction framework. The method is designed to produce climatologically unbiased rainfall fields with correct distributional properties --- adjusting values, aligning distributions, and preserving extremes --- rather than to improve day-to-day prediction skill at the pixel level.

To provide a compact visual summary of these multi-dimensional performance characteristics, Figure 4 presents Taylor diagrams [35] comparing all evaluated products against independent BMKG station observations, stratified by monsoon regime. The diagram simultaneously displays three complementary statistics: Pearson correlation (angular position), normalized standard deviation (radial distance), and centered RMSE (gray dashed contours). A product that perfectly reproduces the observed variability would coincide with the reference point at correlation = 1.0 and normalized standard deviation = 1.0.

The seasonal stratification reveals that the method ordering is stable across both monsoon regimes. In both wet (October--March) and dry (April--September) seasons, LS underestimates the observed variability (median standard deviation ratio of 0.71 and 0.74, respectively), while LSEQM and LSEQM+DL bring the ratio toward unity (1.03/1.00 in the wet season, 1.04/1.02 in the dry season). Correlation remains largely unchanged across correction stages (≈ 0.22 -- 0.25), confirming that distributional correction does not modify temporal prediction skill --- consistent with the theoretical expectation discussed in Section 4.1. CPC-UNI, despite serving as the correction target, does not coincide with the station reference point (standard deviation ratio ≈ 0.85), reflecting the interpolation uncertainty inherent to the gauge analysis at 0.5° resolution --- a characteristic that motivates the independent station validation. IMERG-F, included as an independent benchmark, sits between the uncorrected IMERG-L and the CPC-UNI reference, consistent with its gauge-adjusted nature.

Figure 4: Station-Level Taylor Diagram by Season

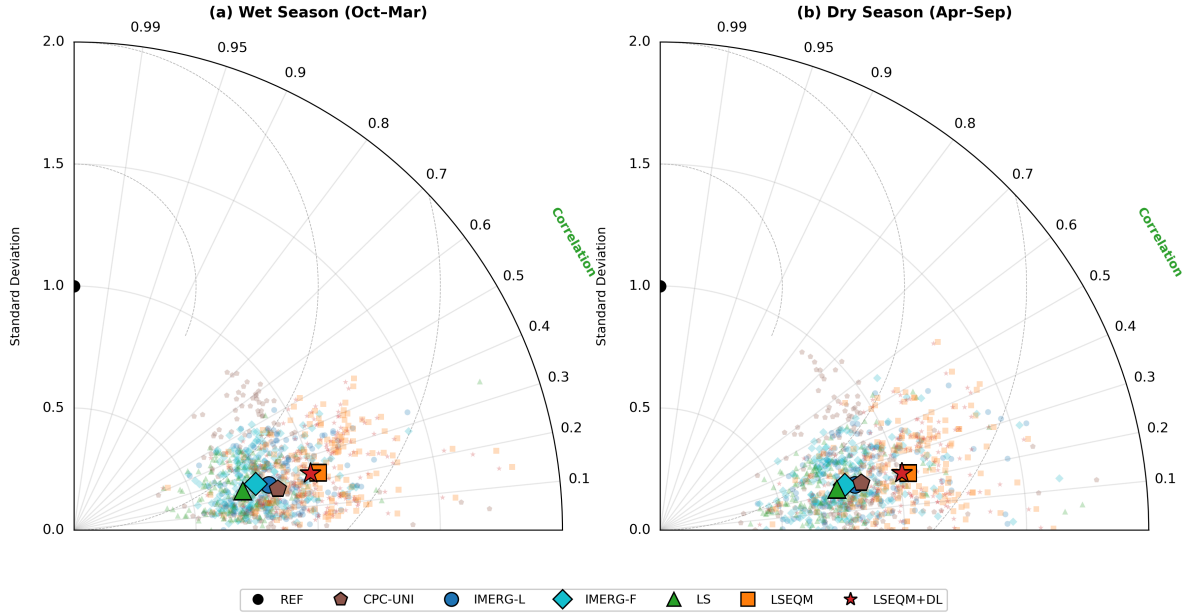


Figure 4. Taylor diagrams comparing daily precipitation performance against independent BMKG station observations, stratified by (a) wet season (October–March) and (b) dry season (April–September). Products shown: CPC-UNI (brown pentagon), IMERG-L (blue circle), IMERG-F (cyan diamond, included as an independent benchmark), LS (green triangle), LSEQM (orange square), and LSEQM+DL (red star). Large markers indicate the median of per-station normalized statistics; small translucent markers show individual stations. The black circle at correlation = 1.0 and normalized standard deviation = 1.0 represents the station reference. Angular position indicates Pearson correlation, radial distance indicates normalized standard deviation, and gray dashed arcs indicate centered RMSE [35]. Statistics are computed from daily paired values across all dekadal periods within each season, for stations with ≥ 30 valid observation days.

The domain-wide averages in Table 4 aggregate regions where the CPC-UNI reference is well constrained by gauge observations alongside station-sparse areas where CPC-UNI itself carries substantial interpolation uncertainty. To disentangle this effect, the analysis was stratified using the station density confidence mask (Section 4.6). In station-covered regions (confidence $C \geq 0.5$, representing 59 of approximately 78,800 land pixels), the CNN refinement effect is clearly visible: the standard deviation ratio improves from 1.16 (LSEQM) to 1.05 (LSEQM+DL), and RMSE decreases from 9.0 to 8.3 mm/day. The relative bias drops from 0.050 (LSEQM) to 0.003 (LSEQM+DL), confirming that the CNN is most effective where the CPC-UNI reference is well constrained. In station-sparse regions ($C < 0.5$), the LSEQM-to-LSEQM+DL transition is minimal, reflecting the confidence mask design that reverts toward purely statistical corrections where few gauges constrain the analysis. This spatial dependence is explored further in Section 4.6.

4.1.1 Conditional Bias Structure: Per-Pixel Quantile Decomposition

The domain-average relative bias reported in Table 4 conceals an important spatial-conditional structure that is not visible in marginal summaries. To expose this, a per-pixel decomposition of the relative bias was performed by partitioning each pixel's time series into three regions defined on the CPC-UNI reference distribution: R1 (body, reference values $\leq Q_{80}$), R2 (moderate tail, $Q_{80} < \text{ref} \leq Q_{99.9}$), and R3 (extreme, $\text{ref} > Q_{99.9}$). The sum-weighted contribution of each region to the overall domain bias is reported as a percentage-point (pp) share of the reference total. Table 4b presents this decomposition for January dekad 1 (peak monsoon) using land pixels only.

Table 4b. Per-pixel R1/R2/R3 decomposition of sum-weighted relative bias against CPC-UNI for January dekad 1, land pixels only. Contributions are expressed in percentage points of the reference sum and are additive to the overall RB.

Field	Overall RB (%)	R1 contrib (pp)	R2 contrib (pp)	R3 contrib (pp)
Raw IMERG-L	+15.52	+38.91	-20.71	-2.68
LSEQM (this study)	+7.90	+31.08	-20.59	-2.60

The decomposition reveals a persistent signature that is conserved across correction stages: IMERG is systematically *too wet* on days that CPC-UNI classifies as light-rain days (R1, +27 to +39 pp) and simultaneously *too dry* on days that CPC-UNI classifies as moderate-heavy or extreme (R2 –21 pp, R3 –2.6 pp). The two sign-opposite errors partially cancel in the marginal summary, but at the pixel-day level they reflect a mismatch in *which days are heavy where* rather than a mismatch in the marginal distribution of daily intensities per pixel. This is a conditional, not a marginal, bias.

A conditional mismatch of this form cannot be removed by univariate quantile mapping by construction, because quantile mapping adjusts the marginal CDF of each pixel without modifying the temporal assignment of values to days. The LSEQM correction visibly flattens R1 (from +38.9 to +31.1 pp) and leaves R2 and R3 essentially unchanged, consistent with a marginal correction acting on an underlying conditional error. The implications of this structure for residual bias attribution are discussed in Section 4.8.5.

4.2 Spatial Distribution of Correction Quality

While domain-averaged statistics provide a useful summary, they mask important spatial heterogeneity in correction performance. Figure 5 presents the Continuous Quality Index (CQI) spatial maps for all three correction stages, computed as the mean CQI across all 36 dekadal periods to provide a climatological summary of correction quality.

The CQI maps reveal a clear west-to-east gradient in correction quality. Java, southern Sumatra, and Bali, regions characterized by dense gauge networks and relatively homogeneous terrain at the CPC-UNI resolution, consistently achieve CQI values in the Fair to Good range (0.5–0.65) across all three methods. Eastern Indonesia, including Papua and Maluku, shows lower CQI values, reflecting both the sparser station coverage of the CPC-UNI reference and the greater topographic complexity of these regions. The CQI ranges are [0.337, 0.657] for LS, [0.283, 0.602] for LSEQM, and [0.285, 0.608] for LSEQM+DL.

The domain-median CQI decreases from 0.541 (LS) to 0.505 (LSEQM) and recovers slightly to 0.508 (LSEQM+DL). This decrease may appear to contradict the distributional improvements documented in Section 4.1, but it reflects the composite nature of CQI: the Temporal Quality Score (TQS), which includes POD and CSI, decreases from 0.666 to 0.648 because the dry-day frequency matching [11] reclassifies some IMERG wet days as dry (reducing POD from 0.77 to 0.71), while the Basic Statistical Quality Score (BSQS) decreases from 0.326 to 0.297 (LSEQM) and 0.299 (LSEQM+DL) due to the increase in RMSE that accompanies the wider corrected distribution. These losses outweigh the gains in Distribution Quality Score (DQS: 0.686 to 0.656 for LSEQM, recovering to 0.661 for LSEQM+DL). The CNN refinement partially recovers performance (LSEQM+DL CQI 0.508 vs LSEQM 0.505), consistent with the targeted spatial adjustment in station-covered regions.

Figure 5: CQI Spatial Distribution by Method

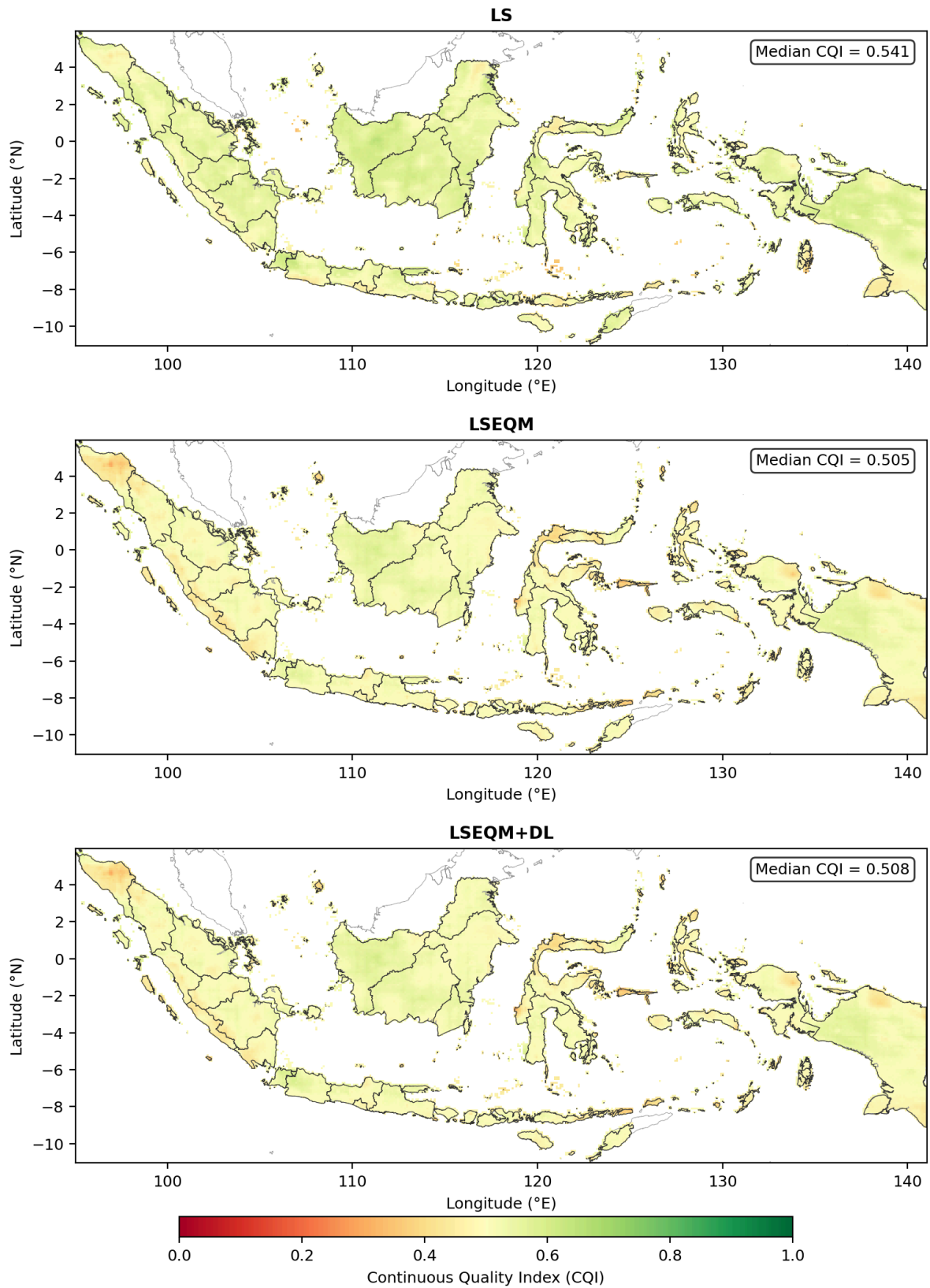


Figure 5. Spatial distribution of the Continuous Quality Index (CQI) for (a) Linear Scaling, (b) LSEQM, and (c) LSEQM+DL, evaluated against CPC-UNI. Values represent the mean CQI across all 36 dekadal periods. Higher values indicate better agreement with the reference dataset. The color scale ranges from 0 (poor) to 1 (excellent).

Figure 6a presents a smoothed CQI difference map (LSEQM+DL minus LS), showing a spatially heterogeneous pattern: regions with higher station density (Java, Bali, southern Sumatra) tend to show modest CQI gains from the CNN refinement, while the domain majority shows small decreases driven by the POD/RMSE trade-off discussed above. Figure 6b presents a six-tier quality classification

for LSEQM+DL that subdivides the dominant Fair range (0.4–0.6) to reveal spatial structure: the majority of land pixels fall in the Fair-High (0.5–0.6) and Fair-Low (0.4–0.5) categories, with a clear west-to-east gradient from Fair-High in station-dense regions to Fair-Low and Marginal (0.3–0.4) in station-sparse eastern Indonesia. Only 2 pixels reach Good (≥ 0.6); no pixels achieve Excellent (≥ 0.7).

Figure 6: CQI Improvement from Hybrid Bias Correction

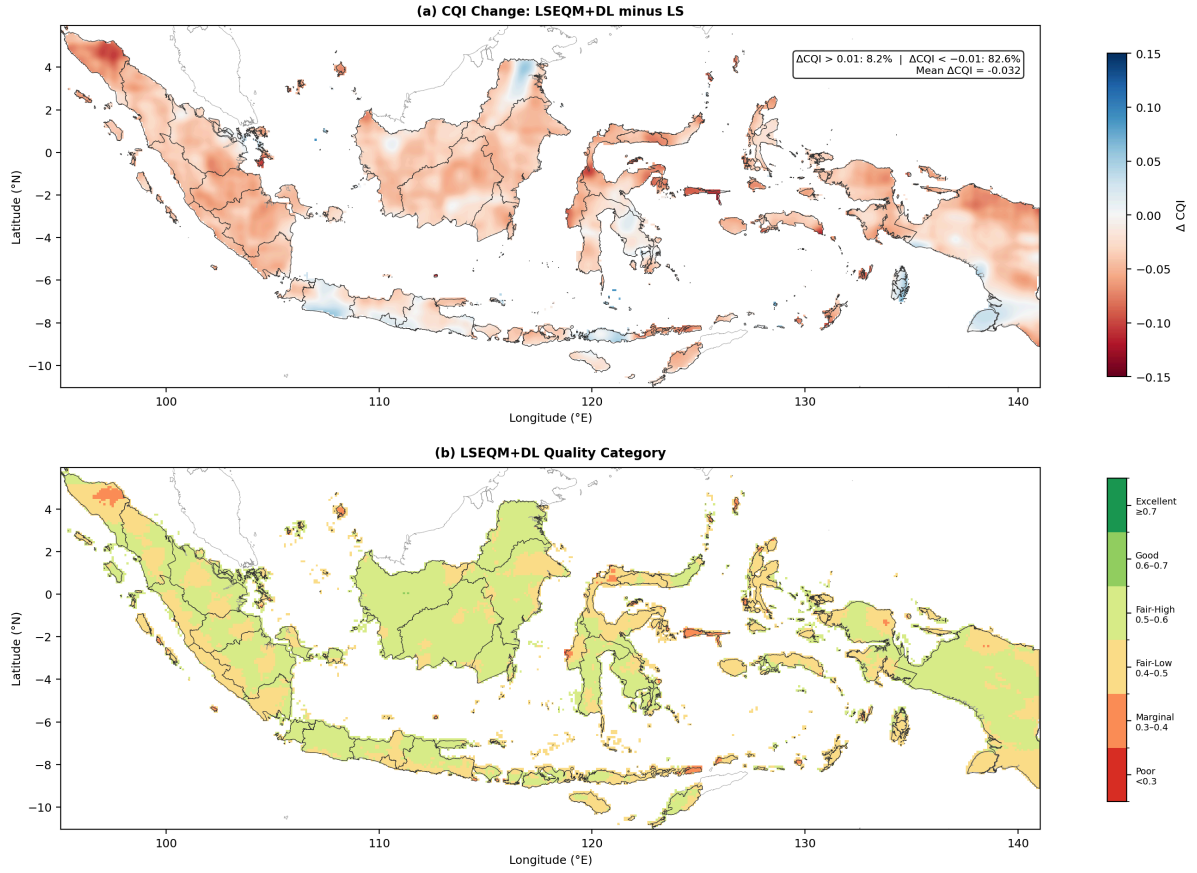


Figure 6. (a) Smoothed CQI difference map showing the mean difference between LSEQM+DL and LS across all 36 dekadal periods (positive values indicate improvement; Gaussian smoothing with $\sigma = 1.5$ pixels applied for visual clarity). (b) Six-tier quality classification of LSEQM+DL: Excellent (≥ 0.7 , dark green), Good (0.6–0.7, green), Fair-High (0.5–0.6, yellow-green), Fair-Low (0.4–0.5, yellow), Marginal (0.3–0.4, orange), and Poor (< 0.3 , red).

The spatial pattern of the CQI difference correlates with station density, a relationship explored further in Section 4.6. The CNN refinement shows its largest positive contribution in station-dense regions, indicating that the alpha-blending strategy and station density confidence mask effectively target the DL contribution where the reference is well constrained. This CQI result highlights an important methodological point: composite quality indices that weight event detection equally with distributional accuracy may undervalue corrections that deliberately trade wet-day frequency for distributional fidelity. The independent station validation (Section 4.5) provides the clearest assessment of the framework's net benefit.

4.3 Distributional Quality and Extreme Event Representation

A central motivation for the hybrid framework is the preservation of extreme precipitation events [46], which are often attenuated by standard bias correction methods. Table 5 presents the domain-median percentile values for the CPC-UNI reference and each correction stage, spanning from the median (Q50) to the 99th percentile (Q99). Percentiles are computed from the single-dekad

aggregated mode, which pools approximately 250 daily values per pixel (25 years $\times$ 10 days per dekad), providing adequate sample sizes for stable quantile estimation.

Table 5. Domain-median precipitation percentiles (mm/day) for the CPC-UNI reference and each correction stage, averaged over all 36 dekadal periods. Values in parentheses indicate the relative error against the reference.

Percentile	CPC-UNI	LS, % Err	LSEQM, % Err	LSEQM+DL, % Err
Q_{25}	0.27	0.44 (+62.6%)	0.00 (−100%)	0.00 (−100%)
Q_{50}	2.22	2.62 (+18.0%)	1.71 (−23.1%)	1.71 (−23.1%)
Q_{75}	8.64	8.84 (+2.4%)	8.76 (+1.4%)	8.76 (+1.4%)
Q_{90}	20.22	19.47 (−3.7%)	22.67 (+12.1%)	22.61 (+11.8%)
Q_{95}	29.80	28.47 (−4.5%)	35.53 (+19.2%)	35.27 (+18.4%)
Q_{99}	51.31	50.59 (−1.4%)	63.19 (+23.2%)	62.46 (+21.7%)

Linear Scaling, which applies a uniform multiplicative factor, preserves the relative percentile structure of the uncorrected satellite field. Because the IMERG-L upper percentiles happen to sit close to the CPC-UNI reference in the domain median once the mean is removed, the LS percentile errors against CPC-UNI appear small at Q_{90} and above. This apparent agreement is a statistical coincidence of the multiplicative rescaling and does not reflect genuine shape correction: LS systematically under-predicts wet-day intensity and over-predicts wet-day frequency (Table 4), and these two errors cancel in the upper percentiles of the pooled sample.

The LSEQM stage modifies the distribution shape through wet-day gamma mapping and explicit dry-day handling. Against the CPC-UNI reference, this produces a positive shift in the upper percentiles (Q_{90} – Q_{99}). The direction of this shift is evaluated more rigorously in Section 4.5 using independent BMKG station observations, where the Q_{99} ratio moves from 0.71 (LS) to 1.05 (LSEQM) and 1.01 (LSEQM+DL) --- that is, the shift away from CPC-UNI is a shift *toward* the gauge ground truth. The residual overshoot observed against CPC-UNI is interpreted in the context of published IMERG-L Maritime Continent wet-bias assessments and CPC-UNI gauge undercatch in Section 4.8.5.

Figure 7 presents the distribution of CQI component scores across all land pixels, comparing the three correction stages. The box plots reveal that all three CQI components decrease from LS to LSEQM, with partial recovery under LSEQM+DL. The BSQS (median: LS 0.326, LSEQM 0.297, LSEQM+DL 0.299) decreases because the wider corrected distribution increases RMSE. The DQS (median: LS 0.686, LSEQM 0.656, LSEQM+DL 0.661) decreases because the CPC-UNI-referenced KS statistic penalizes the intentional distributional shift toward gauge ground truth. The TQS (median: LS 0.666, LSEQM 0.648, LSEQM+DL 0.648) decreases because POD and CSI drop as the dry-day handling [11] reclassifies some IMERG wet days. These component-level patterns explain the overall CQI decrease discussed in Section 4.2 and underscore the tension between CPC-UNI-referenced composite indices and the independent station validation (Section 4.5), which confirms that the distributional corrections represent genuine improvements.

Figure 7: Quality Component Distributions by Method

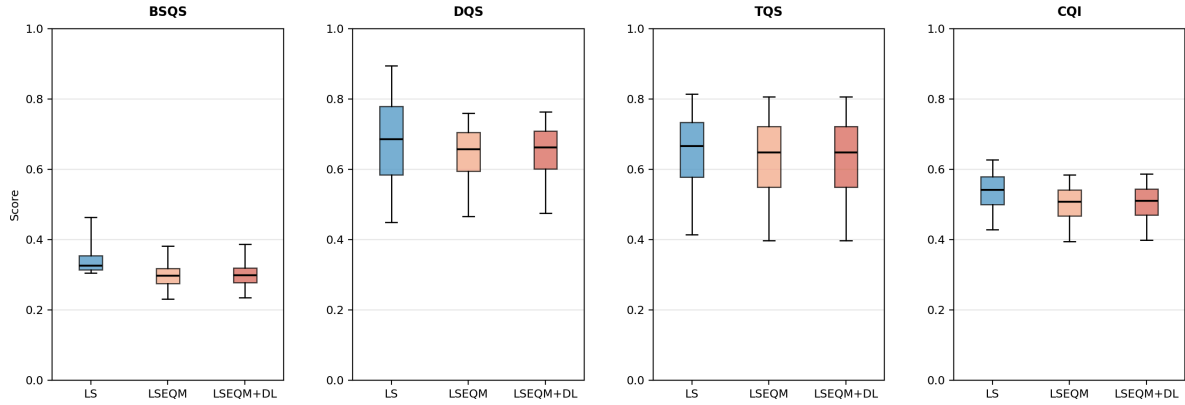


Figure 7. Distribution of CQI component scores across all land pixels for LS (blue), LSEQM (orange), and LSEQM+DL (green): (a) Basic Statistical Quality Score (BSQS), (b) Distribution Quality Score (DQS), (c) Temporal Quality Score (TQS), and (d) overall Continuous Quality Index (CQI). Box plots show the median, interquartile range, and 5th–95th percentile whiskers.

The domain-wide KS p -values present an apparent paradox: the median p -value decreases from 6.56% (LS) to effectively 0.00% (LSEQM and LSEQM+DL), indicating that the corrected distributions are *less* similar to CPC-UNI than the LS product. This is an expected consequence of the EQM design: the correction intentionally shifts the satellite distribution toward independent gauge ground truth (increasing variability and extreme magnitudes), which moves it further from the CPC-UNI reference that underestimates these properties. At independent BMKG stations, the KS p -value improves from 0.01% (LS) to 19.07% (LSEQM and LSEQM+DL), confirming that the distributional shift represents genuine improvement when assessed against uncontaminated ground truth.

4.4 Cross-Reference Evaluation

Evaluating bias-corrected products solely against the training reference (CPC-UNI) risks overly optimistic performance estimates. The evaluation framework includes cross-reference validation against both the IMERG Final Run (IMERG-F), a gauge-adjusted satellite product not used in the correction procedure (Section 3.4.1), and the uncorrected IMERG Late Run (IMERG-L) as a baseline. This implements the 3×3 factorial design described in Table 1, enabling three complementary evaluation perspectives. IMERG-F is also included in the station-level Taylor diagram (Figure 4) as an independent benchmark.

Table 6 presents the full evaluation matrix using metrics aligned with the three-pillar framework. Metrics are summarized as domain-wide medians averaged over all 36 dekadal periods.

Table 6. Cross-reference evaluation matrix organized by the three-pillar framework. RB = Relative Bias (value adjustment), SDR = Std. Dev. Ratio, KS p = Kolmogorov-Smirnov p -value (distribution alignment), $Q_{99}R$ = 99th-percentile ratio (extreme preservation), CSI = Critical Success Index (event detection). Bold indicates the value closest to target per reference group.

Reference	Test	RB	SDR	KS (%)	Ratio	CSI
CPC-UNI	LS	0.0000	0.9707	6.56	0.9669	0.5890
	LSEQM	0.0697	1.1587	0.00	1.2117	0.5723
	LSEQM+DL	0.0652	1.1475	0.00	1.1964	0.5724
IMERG-F	LS	−0.0819	0.9549	51.25	0.9527	0.9016

Reference	Test	RB	SDR	KS p (%)	Q_{99} Ratio	CSI
	LSEQM	— 0.0093	1.1384	0.00	1.2053	0.8574
	LSEQM+DL	—0.0140	1.1274	0.00	1.1902	0.8574
IMERG-L	LS	—0.1010	0.8990	62.26	0.8990	0.9643
	LSEQM	— 0.0279	1.0783	0.00	1.1425	0.9117
	LSEQM+DL	—0.0326	1.0678	0.00	1.1280	0.9116

The CPC-UNI rows confirm the three-pillar pattern documented in Section 4.1: LS zeroes the relative bias by construction, while LSEQM shifts the distribution toward the reference with the Q_{99} ratio moving from 0.97 to 1.21. Against CPC-UNI, LS appears closest to the target for most metrics, because LS preserves the satellite's distributional shape, which happens to approximate the CPC-UNI reference.

The IMERG-F rows provide an independent perspective. Because IMERG-F incorporates gauge data during post-processing, it represents a higher-quality satellite benchmark. The relative bias against IMERG-F decreases substantially from —0.082 (LS) to —0.009 (LSEQM), indicating that the distributional correction brings the product closer to the gauge-adjusted satellite estimate. The high CSI values (0.86–0.90) against IMERG-F reflect the shared satellite retrieval heritage between IMERG-L and IMERG-F. The standard deviation ratio against IMERG-F moves from 0.95 (LS) toward 1.13 (LSEQM+DL), overshooting the IMERG-F variability --- consistent with the interpretation that LSEQM pushes the distribution beyond the gauge-adjusted satellite product and toward independent gauge ground truth (Section 4.5).

The IMERG-L rows quantify the magnitude of correction applied. The relative bias decreases from —0.10 (LS) to —0.03 (LSEQM), and the standard deviation ratio moves from 0.90 toward 1.07, confirming that the framework systematically increases precipitation variability and extreme magnitudes relative to the uncorrected satellite baseline.

A consistent pattern across all three reference groups is that the KS p -value drops to effectively zero after EQM, reflecting the intentional distributional shift introduced by quantile mapping. This shift is confirmed as beneficial by the independent station validation in Section 4.5.

4.5 Station-Level Ground Truth Validation

The gridded evaluation in Sections 4.1–4.4 assesses correction performance against the CPC-UNI reference used during training. The most rigorous test of the framework employs entirely independent ground truth. The corrected products were compared against daily precipitation observations from 171 BMKG (Indonesian Meteorological, Climatological, and Geophysical Agency) weather stations, which were not used in any stage of the bias correction procedure. A minimum of 30 valid overlapping days per station follows WMO guidance for precipitation verification sample sizes; results are robust to increasing this threshold to 60 days (up to 169 stations retained per dekad).

Figure 8 presents the spatial distribution of BMKG stations alongside the per-station standard deviation ratio for the LSEQM+DL product, which measures how well the corrected product reproduces the observed precipitation variability at each gauge location.

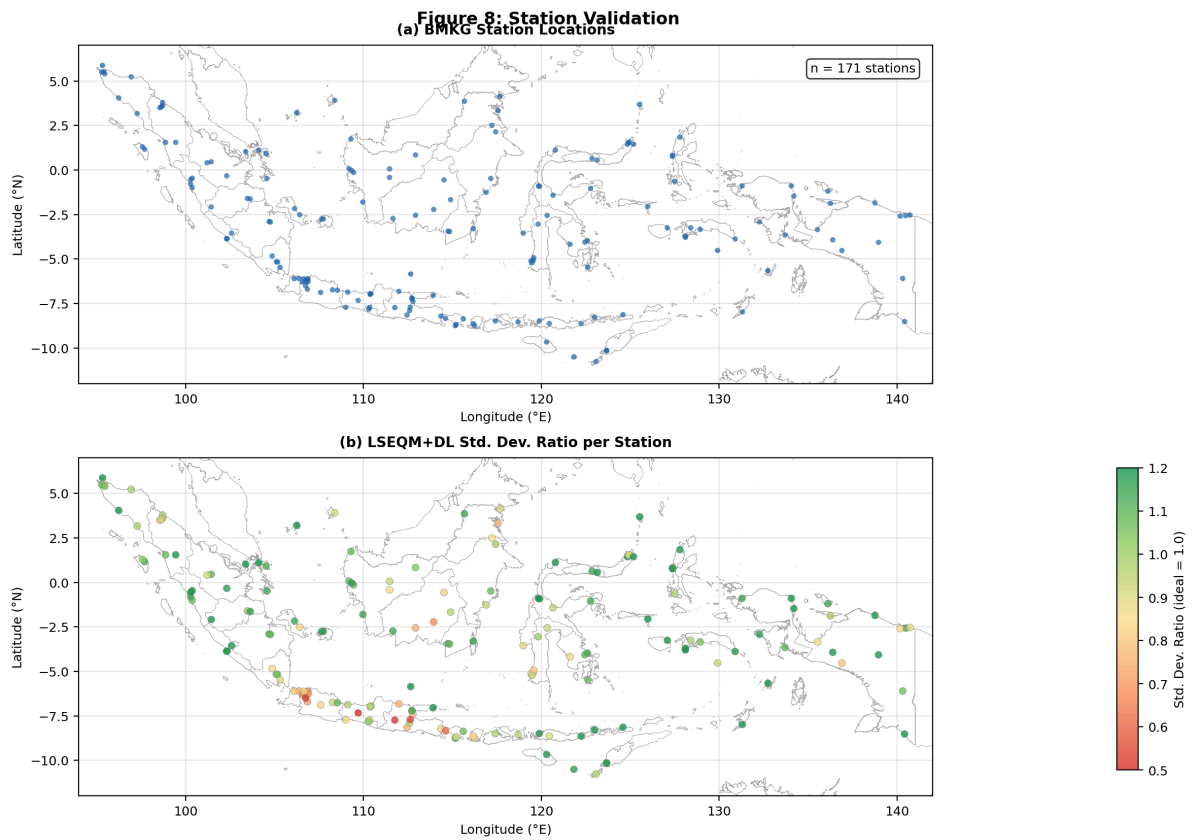


Figure 8. (a) Location of BMKG weather stations used for independent validation ($n = 171$). (b) Per-station standard deviation ratio (test/observed) for the LSEQM+DL product, averaged across all dekadal periods. Values near 1.0 (green) indicate good variability preservation; values below 0.7 (red) indicate underestimation of observed variability.

Table 7 presents the station-level validation results organized by the same three-pillar framework used for the domain-wide evaluation. The station-level results provide the clearest evidence of the framework's effectiveness, as the BMKG observations represent direct ground truth uncontaminated by the interpolation uncertainties present in the gridded CPC-UNI reference.

Table 7. Independent station validation organized by the three design objectives of the hybrid framework. Values represent the median across 171 BMKG stations and all 36 dekadal periods. Ratio metrics express test/observed (target = 1.0). Bold indicates the value closest to the target.

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Value Adjustment	Relative Bias	— 0.114	0.009	−0.006	0
	Std. Dev. Ratio	0.71	1.03	1.00	1.0
Distribution Alignment	KS p -value (%)	0.01	19.07	19.07	100
	Wet-Day Freq. Ratio	1.21	0.96	0.95	1.0
	Mean Wet-Day Precip Ratio	0.71	1.06	1.05	1.0

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Extreme Preservation	Q_{95} Ratio	0.74	1.07	1.05	1.0
	Q_{99} Ratio	0.71	1.05	1.01	1.0
Event Detection	POD	0.78	0.65	0.65	1.0
	CSI	0.53	0.49	0.49	1.0
	FAR	0.36	0.32	0.32	0

The station-level results provide strong evidence for the framework's effectiveness at independent ground truth locations. The most striking changes occur in the LS-to-LSEQM transition. For value adjustment, the standard deviation ratio improves from 0.71 to 1.03 --- meaning that LS underestimates the observed precipitation variability by 29%, while LSEQM matches it within 3%. The LSEQM+DL standard deviation ratio reaches 1.00 (exact match to observed variability). For distribution alignment, the wet-day frequency ratio moves from 1.21 to 0.96, correcting a 21% overestimation of wet-day occurrence to within 5% of observed. For extreme preservation, the Q_{99} ratio improves from 0.71 to 1.05, recovering the large extreme rainfall underestimation present in the LS product; the full LSEQM+DL reaches $Q_{99} = 1.01$ (near perfect). The mean wet-day precipitation ratio improves from 0.71 to 1.06, indicating that the EQM adjustment corrects a systematic 29% underestimation of rainfall intensity on wet days to within 6% of observed values.

These improvements come at a measurable cost in event detection: POD decreases from 0.78 (LS) to 0.65 (LSEQM), while FAR improves from 0.36 to 0.32. This trade-off is inherent to the dry-day frequency matching [11] (Section 3.3), which adjusts the marginal wet-day fraction to match the CPC-UNI reference: some IMERG wet days that correspond to genuine station-observed rainfall are reclassified as dry. However, the net effect is that detected events are more reliable (lower FAR), and the distributional properties of the corrected product are substantially more accurate.

The LSEQM-to-LSEQM+DL transition shows targeted refinement at the station level: the Q_{99} ratio moves from 1.05 to 1.01, and the standard deviation ratio from 1.03 to 1.00 --- both approaching unity. This confirms that the CNN refinement, while conservative by design, provides measurable additional value for extreme event representation.

As with the gridded evaluation, temporal accuracy metrics at stations show no improvement: Pearson correlation remains at approximately 0.24, and RMSE increases from 15.9 (LS) to 18.2 mm/day (LSEQM+DL). This confirms that the framework's value lies in statistical property correction rather than temporal prediction skill, consistent with Section 4.1.

To evaluate extreme event detection skill across a range of precipitation intensities, the WMO-standard multi-threshold verification [24] was applied. Figure 9 presents the Probability of Detection, Critical Success Index, and Equitable Threat Score as functions of precipitation threshold (1, 5, 10, 20, 50, 100, and 150 mm/day).

Figure 9: Multi-Threshold Verification (WMO framework)

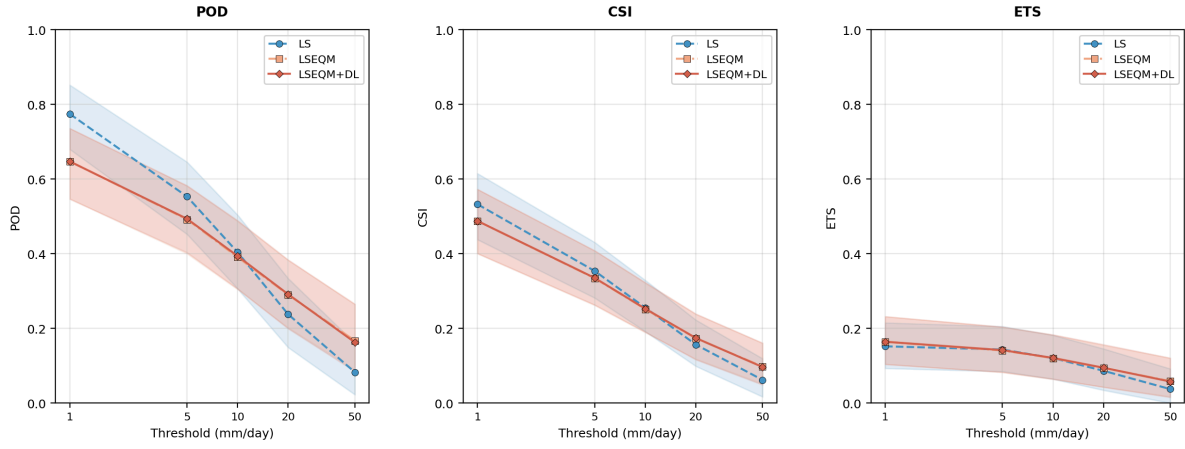


Figure 9. WMO multi-threshold verification: (a) Probability of Detection (POD), (b) Critical Success Index (CSI), and (c) Equitable Threat Score (ETS) as functions of precipitation threshold for LS (blue), LSEQM (orange), and LSEQM+DL (green). Lines show the median across all stations; shading indicates the interquartile range. Thresholds follow WMO classifications: 1 mm (measurable), 5 mm (light), 10 mm (moderate), 20 mm (moderate-heavy), 50 mm (heavy), 100 mm (very heavy), and 150 mm (extreme).

All three methods show the expected decline in detection skill with increasing threshold, as extreme events become rarer and more difficult to reproduce. The LSEQM and LSEQM+DL curves are nearly indistinguishable across all thresholds, reflecting the fact that the CNN refinement adjusts precipitation magnitudes for extreme pixels but does not alter wet/dry classification --- the contingency tables underlying POD, CSI, and ETS are therefore essentially identical for both methods. At the 1 mm threshold, LS achieves the highest POD (0.77) and CSI (0.53), while LSEQM+DL shows lower POD (0.65) but also lower FAR, reflecting the dry-day frequency matching [11] described in Section 3.3. However, at thresholds above 20 mm/day, this pattern reverses: LSEQM and LSEQM+DL maintain higher skill than LS. At the 50 mm threshold, the median CSI for LSEQM+DL is 0.097, compared to 0.062 for LS, representing an improvement of 56%. At 100 mm (very heavy rainfall), the LSEQM+DL product achieves a median POD of 0.065 and CSI of 0.042, compared to LS values of 0.015 and 0.011 respectively, demonstrating that the GPD tail adjustment substantially improves the detection of high-impact events. The Equitable Threat Score, which accounts for random hits, shows a similar pattern: at 50 mm, the ETS for LSEQM+DL (0.058) exceeds that of LS (0.038), confirming that the improved extreme detection is not attributable to increased frequency bias.

These station-level results provide the strongest validation evidence in this study, as BMKG observations are entirely independent of the correction procedure and represent direct ground truth measurements.

4.6 Role of the Station Density Confidence Mask

The station density confidence mask was introduced to manage the spatial heterogeneity of CPC-UNI reference quality by modulating the CNN contribution based on gauge network density. To evaluate the effectiveness of this mechanism, Figure 10a presents the confidence mask itself, derived from BMKG station counts per CPC-UNI grid cell, and Figure 10b examines the relationship between station density and CQI improvement.

Figure 10: Station Density Confidence Mask

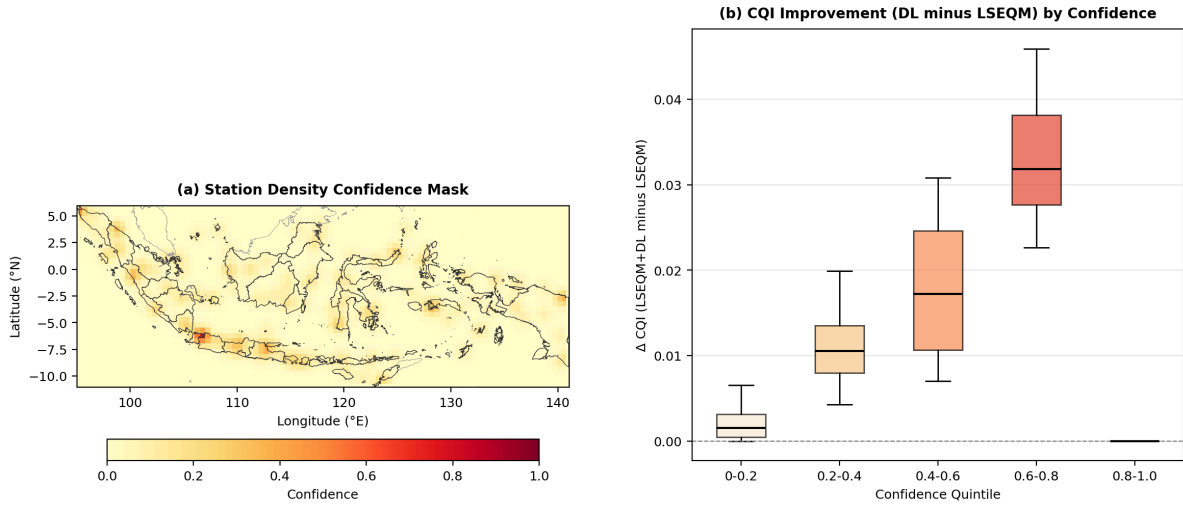


Figure 10. (a) Station density confidence mask $C(x, y)$, ranging from 0 (no station coverage) to 1 (saturated coverage, ≥ 3 stations per 0.5° cell). Gaussian smoothing ($\sigma = 1$ grid cell) prevents sharp boundaries. (b) Relationship between confidence mask value and CQI improvement (LSEQM+DL minus LSEQM): box plots binned by confidence quintile.

The confidence mask reveals the expected pattern: Java, Bali, and parts of Sumatra have high confidence values ($C > 0.7$), while Papua, Maluku, and oceanic boundary regions have near-zero confidence (25.8% of land pixels have $C = 0$; median $C = 0.002$). The relationship between confidence and CQI improvement (Figure 10b) demonstrates that the mask operates as designed: the median ΔCQI (LSEQM+DL minus LSEQM) increases monotonically with confidence, from +0.002 at $C \in [0, 0.2)$ ($n = 18,329$ pixels) to +0.011 at $C \in [0.2, 0.4)$ ($n = 930$), +0.017 at $C \in [0.4, 0.6)$ ($n = 102$), and +0.032 at $C \in [0.6, 0.8)$ ($n = 32$). No pixels reach $C \geq 0.8$, reflecting the inherent sparsity of Indonesia's gauge network. This monotonic relationship confirms that the framework reverts to purely statistical corrections where the training target is unreliable and concentrates CNN refinement where the reference is well constrained.

This behavior has important practical implications. The confidence mask provides users with a spatially explicit quality flag: regions with high confidence values can be considered reliable for climate analysis and risk assessment, while low-confidence regions require additional caution or supplementary validation. For operational agencies such as BMKG, this information supports decision-making about where the corrected product can be used with confidence and where gauge-based assessments remain necessary.

4.7 Temporal Stability of Corrections

An important practical consideration for operational deployment is whether the correction quality remains stable across the annual cycle or varies systematically between wet and dry seasons. Figure 11 presents monthly domain-median values of four key metrics --- standard deviation ratio, RMSE, KS p -value, and CSI --- for all three correction stages.

Figure 11: Monthly Temporal Stability of Corrections

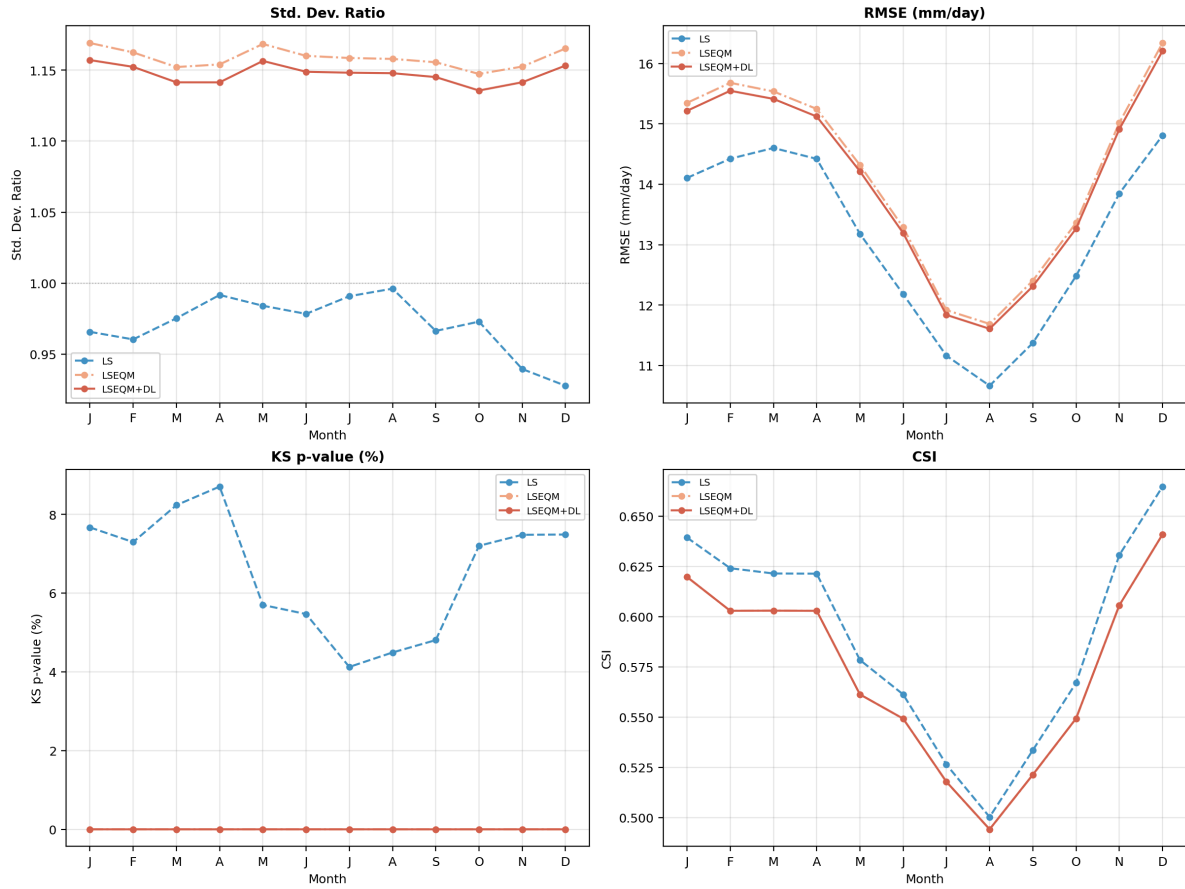


Figure 11. Monthly temporal stability of correction performance: (a) standard deviation ratio, (b) RMSE (mm/day), (c) KS p -value (%), and (d) Critical Success Index (CSI). Lines show domain-median values for LS (blue), LSEQM (orange), and LSEQM+DL (green) across all 12 months. The seasonal cycle reflects the wet (October–March) and dry (April–September) monsoon regime.

The standard deviation ratio (Figure 11a) shows that LSEQM and LSEQM+DL maintain a consistently higher variability than LS across all months, with values ranging from approximately 1.10 to 1.18 compared with 0.94–0.97 for LS. The relative ordering of methods is stable throughout the year, indicating that the distributional correction does not introduce seasonal artifacts. RMSE (Figure 11b) follows the expected seasonal pattern, peaking during the wet season (December–March) when precipitation intensities are highest and declining during the dry season (June–September). LSEQM and LSEQM+DL show moderately higher RMSE than LS, consistent with the wider corrected distribution discussed in Section 4.1.

The KS p -value (Figure 11c) reveals a seasonal pattern for LS, with higher values during the dry season when both satellite and reference distributions are more similar due to reduced precipitation. LSEQM and LSEQM+DL maintain near-zero KS p -values throughout the year, reflecting the intentional distributional shift. CSI (Figure 11d) shows a modest seasonal cycle for all methods, with slightly lower values during the dry season transition months (April–May). The LS-to-LSEQM gap in CSI (approximately 0.02–0.03) is consistent across months, confirming that the POD trade-off discussed in Section 4.5 is not seasonally concentrated.

4.8 Discussion

4.8.1 Synthesis of the Sequential Hybrid Approach

The results presented in Sections 4.1 through 4.6 demonstrate that the sequential LSEQM+DL framework delivers measurable improvements across its three design objectives: adjusting values, aligning distributions, and preserving extremes. Each processing stage addresses a distinct aspect of satellite precipitation bias. Linear Scaling corrects the first moment (mean bias). EQM aligns the full marginal distribution, bringing the wet-day frequency, variability, and moderate percentiles into agreement with the reference. GPD tail adjustment preserves extreme event magnitudes that standard EQM attenuates due to limited sampling in the upper tail. The CNN refinement provides spatial consistency by learning residual patterns from the corrected fields. The dual-resolution fitting strategy, which estimates CPC-UNI distribution parameters at the native 0.5° grid and interpolates them smoothly to the 0.1° IMERG grid (Section 3.3), produces spatially continuous correction fields consistent with the BCSD principle [23].

The clearest demonstration of this sequential design comes from the independent station validation (Table 7), where each pillar shows substantial improvement from LS to LSEQM: the standard deviation ratio improves from 0.71 to 1.03 (and to 1.00 under LSEQM+DL), the Q_{99} ratio from 0.71 to 1.05 (1.01 under LSEQM+DL), and the wet-day frequency bias decreases from 21% to 5%. These improvements at independent ground truth locations confirm that the framework addresses genuine precipitation biases rather than merely fitting the training reference.

An equally important finding is that distributional bias correction does not improve --- and is not designed to improve --- temporal prediction skill at the pixel level. Correlation, RMSE, and NSE remain largely unchanged across correction stages (Section 4.1), consistent with the theoretical understanding that quantile mapping adjusts marginal distributions without modifying the temporal ordering of values [34]. This distinction is critical for users: the corrected product should be interpreted as having improved statistical properties (bias, variability, distribution shape, extreme magnitudes) rather than improved day-to-day prediction accuracy. Applications that require temporally accurate fields, such as real-time flood forecasting, should use the corrected product in combination with temporal downscaling or ensemble methods.

This sequential design offers advantages over both purely statistical and purely DL approaches. Single-method statistical corrections can align distributions but may not capture complex spatial patterns driven by topography and atmospheric processes [8,44]. Standalone DL approaches can learn spatial relationships but risk introducing physically implausible values if not adequately constrained [9,12]. By positioning the CNN as a refinement step after statistical correction, the DL component operates on partially corrected input from the same statistical domain used during training, avoiding the domain shift problem that can degrade purely DL methods.

The alpha-blending strategy further constrains the DL contribution by design: the base weight of 0.70 assigns dominant influence to the statistical correction, while the CNN provides targeted refinement for extreme pixels only. This conservative integration reflects the principle that statistical methods with established theoretical foundations should anchor the correction, with machine learning serving as a complementary tool rather than a replacement.

It should also be noted that IMERG-L is known to over-detect light and drizzle-class precipitation over the tropical oceans and coastal zones, a characteristic consistently reported in independent evaluations [31,32]. The R1 contribution identified in Section 4.1.1 is consistent with this well-documented light-rain over-detection. Because LSEQM operates pixel-wise on marginal distributions, it partially flattens but does not eliminate this signature, which is revisited in Section 4.8.5.

4.8.2 Extreme Event Representation

The preservation of extreme precipitation events emerged as a key strength of the framework. The GPD tail adjustment at the 80th percentile threshold addresses a well-documented limitation of standard EQM: the tendency to attenuate extreme values due to insufficient sampling in the upper tail of the empirical distribution [11]. By modeling exceedances explicitly and employing K-fold cross-validation for parameter estimation, the GPD stage provides stable extreme quantile corrections even in regions with limited observational samples.

The WMO multi-threshold verification (Figure 9) provides strong quantitative evidence, as it evaluates detection skill across the full precipitation intensity spectrum using operationally defined thresholds. The maintained skill at thresholds above 50 mm/day is critical for applications in flood early warning and urban drainage design, where accurate representation of heavy and extreme rainfall directly impacts risk assessment and emergency response planning.

4.8.3 Practical Implications of the Confidence Mask

The station density confidence mask addresses a challenge common to all bias correction methods that rely on gridded gauge products: the spatial heterogeneity of reference data quality. As demonstrated in Section 4.6, the mask modulates the DL influence based on the density of contributing stations, preventing CNN artifacts in poorly observed regions.

A practical advantage of this design is its transferability. The mask adapts automatically to any station network, requiring no manual tuning of regional parameters. For regions with denser networks (e.g., parts of India, Japan), the mask would permit greater DL influence, while regions with sparser coverage (e.g., Central Africa, the Amazon basin) would receive predominantly statistical corrections. The mask also serves as a spatially explicit quality indicator for end users, supporting operational decisions about where the corrected product can be applied for climate analysis, hydrological modeling, or risk assessment.

4.8.4 Limitations

Several limitations should be acknowledged. First, the bias correction is performed separately for each dekadal period, pooling daily values across years. This design preserves seasonal characteristics and provides adequate sample sizes for distribution fitting, but does not explicitly model temporal autocorrelation within each dekad or inter-annual trends in precipitation patterns.

Second, the CPC-UNI reference product is available at 0.5° spatial resolution, considerably coarser than the 0.1° IMERG grid. The dual-resolution fitting strategy (Section 3.3) produces spatially smooth correction fields by interpolating fitted parameters, but the correction still targets the spatial scales resolved by CPC-UNI rather than the finer scales present in IMERG. Bilinear interpolation of parameters does not introduce sub-grid variability. Consequently, the framework corrects the coarse-scale distributional properties of the precipitation field without adding fine-scale detail in regions of complex topography.

Third, the CNN architecture employs a Flatten-Dense bottleneck that compresses spatial information into a one-dimensional representation before reconstruction. This relatively simple architecture was selected deliberately: because the CNN operates as a refinement step on already-corrected fields rather than as the primary correction, a lightweight model reduces overfitting risk and training cost. Nevertheless, fully convolutional architectures (e.g., U-Net) that preserve spatial resolution throughout the network could improve performance and would be worth exploring in future work.

Fourth, this study demonstrates the framework over a single domain (Indonesia) and a single temporal period (2001–2025 for IMERG Late Run and CPC-UNI). While the methodology relies on globally available datasets (IMERG and CPC-UNI) and employs transferable statistical methods, the optimal GPD threshold, CNN hyperparameters, and alpha-blending weight may require recalibration for different climate regimes [44,45]. Performance varies between the wet season (October–March) and dry season (April–September), as detailed in Section 4.7.

Fifth, the evaluation uses the same temporal period for both correction and assessment. A strict temporal hold-out (e.g., training on 2001–2020 and evaluating on 2021–2025) would provide a more rigorous assessment of generalization, though at the cost of reducing the training sample from approximately 250 to 200 daily values per dekad, which may compromise GPD parameter stability. The use of independent validation against BMKG stations (Section 4.5) partially mitigates this concern.

Sixth, the alpha-blending weight ($\alpha = 0.70$) and GPD threshold percentile (80th) were selected based on domain expertise rather than formal optimization. A sensitivity analysis varying these parameters could identify performance trade-offs and inform region-specific calibration. Similarly, the QA framework component weights (basic statistical quality: 0.35, distribution quality: 0.35, temporal quality: 0.30) represent a subjective but informed partitioning of quality dimensions; alternative weightings could shift the CQI values and categorical classifications, though the relative ranking of correction methods is expected to be robust to reasonable weight perturbations.

4.8.5 Residual Bias Attribution

After LSEQM correction, IMERG-L exhibits a residual positive bias of approximately +8% against CPC-UNI over Indonesia during peak monsoon (January dekad 1, land pixels only), compared with +15.5% for the raw IMERG-L product. This residual persists across daily and dekadal evaluation framings and across alternative tail-handling configurations (parametric GPD, empirical tail, frequency-matched drizzle suppression), indicating a structural rather than methodological origin.

Two factors inherent to the CPC-UNI reference explain the floor. First, CPC-UNI systematically undercatches convective precipitation over sparsely gauged mountainous terrain in the tropics, a well-documented limitation of gauge-based interpolation products [36,37]. Second, the 0.5° gauge analysis cannot resolve sub-grid heterogeneity that is physically present in the 0.1° satellite retrieval, so a portion of the IMERG wet signal that is locally correct at 0.1° is unrecoverable at the CPC-UNI resolution regardless of statistical treatment.

The magnitude of the residual is consistent with published independent evaluations of IMERG-L over the Maritime Continent. Several studies using diverse reference datasets [31,32,33,38,39,47] report satellite wet biases in the range of +10% to +25% during the wet season for this region. The +8% residual reported here after LSEQM therefore falls at the lower end of the published wet-season range.

The per-pixel R1/R2/R3 decomposition (Section 4.1.1) further shows that the residual is conditional in nature: IMERG is too wet on CPC light-rain days ($R1 \approx +31$ pp) and simultaneously too dry on CPC moderate-to-extreme days ($R2 \approx -21$ pp, $R3 \approx -2.6$ pp), and these sign-opposite errors partially cancel in the domain summary. Because marginal quantile mapping adjusts each pixel's CDF without reassigning values to days, a conditional mismatch of this form cannot be removed by LSEQM or by any univariate bias correction method by construction [34,11]. It reflects a mismatch in *which days are heavy where* between IMERG and the gauge analysis, not a mismatch in the marginal distribution of daily intensities.

Taken together, these observations support interpreting the post-LSEQM residual as reference-side uncertainty rather than correction deficiency. The station-level validation against BMKG observations (Section 4.5) provides the independent ground-truth perspective required to resolve this interpretation, and shows that LSEQM moves the standard deviation ratio, wet-day frequency, and Q_{99} toward the gauge observations even where it moves away from the gridded CPC-UNI reference.

4.8.6 Future Directions

Several extensions of this work merit investigation. Replacing the Dense reconstruction layer with a fully convolutional architecture (e.g., U-Net) could improve the spatial fidelity of the CNN refinement [40,41], particularly over regions with sharp topographic gradients. Incorporating additional predictive features, such as elevation, land cover type, and distance to coast, as auxiliary CNN input channels could provide the model with physically meaningful spatial context beyond the precipitation field alone.

Extending the framework to other satellite precipitation products (e.g., CHIRPS, GSMaP, MSWEP [42]) and other correction-reference pairings would test the generalizability of the sequential approach. Application to other tropical and subtropical regions with sparse gauge networks, such as Sub-Saharan Africa and the Amazon basin, would directly assess the transferability claim.

At the temporal scale, extending the framework from dekadal to pentad (5-day) or individual daily corrections would increase the temporal specificity of the bias correction, though at the cost of reduced sample sizes per fitting window. Adaptive window sizing, where the pooling period is adjusted based on data availability, could balance temporal specificity against statistical robustness.

A formal sensitivity analysis of the alpha-blending weight and GPD threshold percentile, potentially using cross-validated grid search [43], would establish whether the current parameter choices are stable across different climate regimes or require region-specific tuning. Similarly, a strict temporal hold-out experiment would strengthen the generalization claims, and is planned as part of the operational deployment of this framework.

Finally, the framework could be extended to address additional precipitation characteristics beyond daily accumulations, such as sub-daily intensity, the diurnal cycle, or multi-day event persistence, which are important for hydrological applications but are not explicitly targeted by the current dekadal-based correction design.

Conclusions

This study developed and evaluated a hybrid bias correction framework (LSEQM+DL) for daily satellite precipitation that sequentially applies Linear Scaling, Empirical Quantile Mapping with Generalized Pareto Distribution (GPD) tail adjustment, and Convolutional Neural Network (CNN) refinement. Applied to IMERG Late Run daily precipitation over Indonesia (2001–2025) and evaluated against CPC-UNI and 171 independent BMKG station observations, the framework yields four principal findings, organized around its three design objectives: adjusting values, aligning distributions, and preserving extremes.

First, the sequential design delivers measurable improvements across all three objectives. At independent BMKG station locations, the standard deviation ratio improves from 0.71 (LS) to 1.03 (LSEQM) and 1.00 (LSEQM+DL), the wet-day frequency bias decreases from 21% to 5%, and the Q_{99} ratio improves from 0.71 to 1.05 (LSEQM) and 1.01 (LSEQM+DL), recovering the extreme rainfall underestimation present in the LS product. These improvements come at a measurable cost in event detection (POD decreases from 0.78 to 0.65), a trade-off inherent to the dry-day frequency matching [11] that is partially offset by improved FAR (0.36 to 0.32). Each correction stage targets a

distinct bias dimension: LS adjusts the mean, EQM aligns the distribution, and GPD preserves the extremes.

Second, the framework honestly operates within the known limitations of distributional bias correction: pixel-level temporal accuracy (correlation, RMSE, NSE) remains largely unchanged across correction stages, consistent with the theoretical understanding that quantile mapping adjusts marginal distributions without modifying temporal sequencing [34]. This distinction is important for users: the corrected product has improved statistical properties (bias, variability, distribution shape, extreme magnitudes) but not improved day-to-day prediction accuracy.

Third, independent validation against 171 BMKG stations strengthens confidence that the improvements reflect genuine enhancement of the precipitation signal rather than overfitting to the CPC-UNI training reference. The domain-wide CQI decreases from 0.541 (LS) to 0.508 (LSEQM+DL), driven by the POD trade-off in the composite index; however, the station-level validation confirms that the distributional corrections move the product toward independent ground truth. The WMO multi-threshold verification further demonstrates that at thresholds above 20 mm/day, LSEQM+DL maintains higher detection skill than LS, with the CSI at 50 mm improving by 56%.

Fourth, the station density confidence mask provides a practical mechanism for managing reference dataset uncertainty. By modulating the CNN influence based on gauge network density, the mask prevents DL artifacts in station-sparse regions while allowing CNN refinement where the reference is well constrained. The mask also serves as a spatially explicit quality flag for end users.

Several limitations warrant consideration, including the absence of a strict temporal hold-out, the use of a relatively simple CNN architecture, and the reliance on fixed values for the blending weight and GPD threshold that were not formally optimized. These choices were guided by practical constraints of limited training samples per dekad (~230 daily values) and the design priority of maintaining physical plausibility over maximizing statistical fit. Future work should address these limitations through sensitivity analysis, alternative architectures such as U-Net, and application to additional tropical regions to test transferability.

The framework relies on globally available datasets (IMERG and CPC-UNI) and employs transferable statistical-distributional methods, making it applicable to other regions with sparse gauge networks. The modular implementation supports integration into operational precipitation monitoring workflows where climatologically unbiased daily estimates are needed for hydrological modeling, agricultural planning, and climate risk assessment.

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